Emotional Data Model (EDM) v0.8.3

A protocol-level representation and persistence architecture for governed emotional data.

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Abstract

Artificial intelligence systems currently lack a stable, interoperable substrate for representing affective context. While contemporary large language models internalise emotional, narrative, and motivational cues transiently within latent activations, these signals remain non-portable, non-inspectable, and vulnerable to drift across model updates, vendors, and computational environments. The **Emotional Data Model (EDM)** addresses this gap by defining a governed, schema-bound representational format that externalises affective context as a deterministic, model-agnostic data object. EDM v0.8.3 specifies domain-complete structures for narrative anchors, affective topology, salience geometry, contextual framing, and in-band governance, ensuring long-range interpretability without reliance on inference or behavioural semantics.

To ensure persistence, transportability, and sovereignty, EDM artifacts are encapsulated within the **.ddna envelope**, a byte-stable container providing canonical serialisation, integrity guarantees, and immutable governance attachment. The combined architecture establishes a clear separation between **Representation**, **Persistence**, and **Computation**, enabling downstream systems to compute over stable artifacts without redefining their structure. This whitepaper formalises the representational architecture, versioning discipline, and persistence model required for durable affective context in AI ecosystems, establishing a foundation for **sovereign emotional memory** capable of supporting cross-vendor interoperability and coherence-aligned computation across future model generations.

Core Contribution

This work formalises the distinction between semantic content and affective state, proposing a composite representational geometry defined as:

$$\mathbf{R(x) = (\Sigma(x), E(x))}$$

where **Σ(x)** denotes the semantic substrate and **E(x)** denotes the structured affective mapping encoded via the EDM schema. Unlike current approaches, where **E(x)** is implicit and transient, this protocol instantiates **E(x)** as an explicit, governed, and persistent data object.

By externalising these elements, the protocol achieves three architectural invariants:

1. **Representational Stability:** Emotional signals remain interpretable across model families and computational eras, decoupled from the specific inference engine used to generate them.
2. **Sovereign Persistence:** The .ddna envelope binds governance logic—including jurisdiction, identity rights, and retention policies—directly to the data payload, ensuring compliance persists during transport.
3. **Non-Inferential Structure:** The architecture enforces a strict boundary where representation precedes computation, supporting the emergence of coherence-aligned AI systems built on stable data rather than implicit behaviour.

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0. Executive Architecture Summary

The Emotional Data Model (EDM) v0.8.3 defines a governed, schema-bound representational architecture for encoding affective context as a stable, machine-interpretable data object. The architecture specifies the complete set of structural domains, serialisation rules, versioning guarantees, and in-band governance requirements necessary to ensure representational fidelity across heterogeneous systems.

The schema defines a representational manifold of ten canonical domains — `meta`, `core`, `constellation`, `milky_way`, `gravity`, `impulse`, `governance`, `telemetry`, `system`, and `crosswalks` — plus an optional `extensions` domain for partner-namespaced enrichments. Three implementation profiles (Essential, Extended, Full) determine which domains and fields are active for a given artifact, with partner profiles enabling vertical-specific field selection within that structure.

Representation is preserved by the **.ddna envelope**, which functions as the architecture's persistence and sovereignty layer. The .ddna structure encapsulates the unmodified EDM payload, enforces byte-stable serialisation, binds governance and jurisdictional metadata, and provides integrity and provenance guarantees through deterministic hashing and optional signing. The envelope is strictly prohibited from introducing behavioural semantics, altering representational content, reinterpreting schema fields, or embedding model-dependent structures.

To reconcile immediate utility with long-term sovereignty, the architecture supports two operational modalities: a **stateless configuration** for immediate session-level coherence, where the artifact remains ephemeral and unsealed; and a **persistent configuration**, where the artifact is sealed within the .ddna envelope to ensure sovereignty, rights preservation, and transportability.

Together, EDM and .ddna form the **Emotional Data Architecture Stack**: a representational layer and a persistence layer that operate independently yet jointly enforce structural invariants, governance fidelity, and long-term interpretability. This architecture provides the conditions under which affective context can exist as a versioned, governed, interoperable, and platform-neutral data object.

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1. Architectural Foundations of EDM v0.8.3

1.1 The Representational Deficit

Contemporary intelligent systems lack an external, governed, and durable representational layer for affective context. While model-internal activations may correlate with affective features, these signals are transient, architecture-bound, and non-deterministic. The implicit signal definitions of these states drift with training cycles, architectural revisions, sampling behaviour, and vendor-specific implementations, preventing the formation of consistent, cross-model referents for emotional structure. This absence renders affective context non-transferable and computationally brittle, as downstream systems must rely on unstable latent geometries rather than fixed representational coordinates.

1.2 The Sovereignty Gap

Beyond technical instability, the lack of an explicit schema creates a fundamental safety and compliance vulnerability. Because internal emotional states are not discrete data objects, they cannot encode retention policies, jurisdictional constraints, or subject rights. An inferred emotional state exists only as a probability within a neural network, making it inherently ungovernable. It cannot be audited for bias, deleted upon request, or restricted by consent. EDM addresses this deficiency by externalising affective context into a structured, domain-complete object. By transforming implicit behaviour into explicit data, the architecture enables the attachment of in-band governance, ensuring that emotional signals are not only interpretable but legally and ethically compliant across their lifecycle.

1.3 Architectural Principles

The EDM architecture is governed by three non-negotiable normative principles designed to ensure long-range stability and safety:

- **Explicit Representational Structure:** All affective contexts **MUST** be encoded as explicit fields within a declared profile schema. No diagnosed, reconstructed, or predictive content is permitted to enter the representational payload; population **MUST** be derived from explicit extraction or direct declaration, ensuring the artifact records **extracted structure** rather than **clinical diagnosis** or **behavioural prediction**.
- **In-Band Governance:** Governance descriptors—including consent basis, jurisdiction, retention, visibility, exportability, policy constraints, and rights declarations—**MUST** be encoded in-band within the representational layer. Governance is structurally inseparable from the artifact and cannot be detached, overridden, or externally appended.
- **Strict Layer Separation:** Representation, persistence, and computation act as distinct architectural layers. EDM v0.8.3 defines representational structure only; the .ddna envelope defines persistence and sovereignty only; external systems define computational behaviour only. No layer may redefine the **representational semantics** or geometry of another. Structural meaning **MUST NOT** depend on model architecture, embedding space, inference strategy, or downstream computational operator.

1.4 Scope

This specification defines the **Emotional Data Architecture Stack**. Its normative scope includes:

1. **EDM v0.8.3:** The governed, domain-complete representational schema for affective context.
2. **The .ddna Envelope:** The sovereign, byte-stable persistence and integrity layer.
3. **The Compute Boundary:** The interface definitions required to expose artifacts to the External Compute Layer.

The following are **explicitly out of scope** for this specification: the internal logic of inference processes, appraisal mechanisms, behavioural modelling, psychological interpretation, optimisation strategies, and the algorithms of **downstream coherence engines**. These belong exclusively to the External Compute Layer and do not form part of the representational or persistence definitions.

2. Background & Prior Work

The Emotional Data Architecture Stack addresses the absence of an explicit, governed, and structurally stable representation of affective context in contemporary AI systems. Existing architectures provide mechanisms for linguistic transformation, pattern modelling, and multimodal inference, but they do not provide a data-level substrate capable of encoding affective structure as a durable, interpretable, and transportable object. This section situates EDM v0.8.3 and the .ddna envelope within the technical lineage that motivates the need for an externalised representational layer.

2.1 Affective Context as the Organising Substrate

In biological intelligence, affective context functions as the organising substrate of memory and attention. Research in affective neuroscience (Damasio, Panksepp) demonstrates that emotion is not a post-hoc reaction to processing, but a **pre-computational substrate** that constrains search spaces, assigns salience, and ensures coherence. In these systems, the "Somatic Marker"—a stable, felt state—precedes and guides high-level reasoning. This reference is included solely for historical lineage; EDM does not implement somatic-marker theory nor any biological or behavioural mechanisms, and the concept is not imported into the representational schema.

Contemporary AI architectures invert this priority. They attempt to derive affective context as a transient byproduct of semantic processing, rather than establishing it as a foundational input. By treating emotion as a latent output rather than a structural input, current systems suffer from a **representational inversion**: they attempt to compute **significance** without a stable reference for **what is significant**. EDM v0.8.3 resolves this inversion by re-establishing the biological order: it defines a stable, governed representational layer for affect that exists *before* and *independent of* the reasoning and inference process.

2.2 Limitations of Model-Internal Signals

Model-internal affective signals fail as representational structures because they are artefacts of inference rather than explicitly defined data. Their formation depends on vendor-specific training regimes, latent geometries, and architectural parameters that lack cross-model continuity. These internal signals exhibit **architectural instability**, as internal feature representations are specific to a model family and cannot be decoded or interpreted by unrelated architectures. They demonstrate **temporal instability**, where model updates, fine-tunes, and retraining cycles alter internal geometries, preventing long-range continuity. **Contextual variability** arises where input phrasing, sampling behaviour, and runtime conditions alter internal activations in non-deterministic ways. Finally, they suffer from **governance absence**, as internal activations cannot encode rights, jurisdiction, retention policy, or consent metadata. Because model-internal structures cannot be stabilised, governed, or externalised, they do not constitute a viable substrate for representational architecture.

2.3 Structural Requirements for Durable Representation

To establish a durable, affective representation layer, the architecture must satisfy specific structural requirements. The representation must be **Explicit**, encoding affective structures as concrete schema fields rather than latent vectors. It must enforce **Profile Completeness**, ensuring all domains and fields defined by the declared Implementation Profile are present, and that no domains or fields outside the profile definition are included. **Representational Semantics** must be **Non-Inferential:** fields are populated through extraction from expressed source content — including LLM-assisted interpretation of that content — but **MUST NOT** introduce latent reconstruction, psychological inference, or predictive content not grounded in the source. Furthermore, the architecture requires **In-Band Governance**, where rights and obligations are encoded in the object itself, and **Persistence**, ensuring artifacts remain byte-stable and portable across vendors and time. Finally, the schema must maintain **Model Neutrality**, independent of any embedding family or internal geometry.

2.4 Technical Lineage

EDM v0.8.3 draws architectural precedent from established representational systems without inheriting their specific semantics. It builds upon the tradition of structured augmentation systems established by Bush, Engelbart, and Berners-Lee, which demonstrated the necessity of addressable, governed data layers. It also aligns with schema-first interoperability frameworks such as JSON-LD and HL7 FHIR, which establish that governed, domain-complete schemas are essential for long-range interoperability across heterogeneous vendors. Crucially, the protocol's sovereignty model draws architectural inspiration from W3C Verifiable Credentials (VCs) and Decentralized Identifiers (DIDs), adopting the principle of cryptographically separating the subject (User) from the issuer (System) to ensure portable, user-centric control over sensitive claims.

2.5 Structural Gaps in Existing Memory Systems

Despite these precedents, existing memory systems in AI—including vector stores, RAG pipelines, and proprietary caches—lack essential representational capabilities for affect. They fail to provide explicit affective representation, rights-bearing metadata, persistent salience descriptors, or transportable interpretability. EDM addresses these gaps by defining a governed, schema-complete, model-neutral representational **significance** substrate that can persist independently of any computational system.

3. Architectural Overview

The EDM architecture establishes the representational substrate required for the stable, governed encoding of affective context within intelligent systems. The architecture functions exclusively as a representational specification; it does not embed computational logic, behavioural operators, or inferential processes. Instead, it specifies the representational and persistence layers that form the foundation upon which independent computational systems may operate. This separation ensures that affective context remains a consistent, model-neutral object of interpretation, insulated from changes in inference techniques, platform implementations, and vendor ecosystems.

All computation—including appraisal, retrieval, weighting, generation, coherence operations, or alignment—remains strictly outside the representational layer. EDM defines the structural form of affective context; the .ddna envelope defines its preservation mechanism; and external systems determine its usage. This division is fundamental to the architecture's long-term stability.

3.1 High-Level Design Goals & Constraints

The architecture is guided by normative design constraints that define its structural boundaries and ensure long-term viability.

- **Representational Stability:** Semantics must remain fixed regardless of changes in model architecture, embedding methods, versioning, or platform transitions.
- **Interpretive Consistency:** The structural definition of every field is defined exclusively by the schema, ensuring a stable representational contract across heterogeneous systems without enforcing identical subjective interpretation.
- **Governance Integration:** Rights, consent, retention, jurisdiction, visibility, and exportability constraints are encoded in-band and may not be detached, overridden, or altered by downstream actors.
- **Architecture Neutrality:** The schema must not depend on embeddings, inference systems, latent geometries, or model-internal patterns.
- **Persistence Durability:** Artifacts must retain byte-stable identity across storage, transport, and system migration.
- **Non-Inferential Representation:** Only explicit, extracted, or declared information may appear in representational fields; no psychological reconstruction or latent inference is permitted.
- **Interoperability:** Achieved through crosswalk mappings that reference both established taxonomies and emerging sensing modalities—including prosody, facial landmarks, and empathetic inference signals—without altering domain semantics.

3.2 The EDM Manifold

The EDM schema defines a representational manifold composed of five interdependent central domains — Core, Constellation, Milky_Way, Gravity, and Impulse — supported by five peripheral governance and metadata structures: Meta, Governance, Telemetry, System, Crosswalks, and an optional Extensions domain for partner-namespaced enrichments. Each domain contributes a distinct structural dimension of affective context. The active domain set for any artifact is determined by its declared implementation profile (Section 3.7).

The `Core` domain establishes narrative anchoring through structural primitives that define the internal organisation of the context, while `Constellation` encodes affective topology using explicit, non-inferential fields. The `Milky_Way` domain captures contextual placement and environmental markers, whereas `Gravity` defines salience parameters and recurrence structure. `Impulse` records motivational and directional configuration as explicit state vectors without behavioural inference. These domains function as a complete set within the Full profile; they **MUST NOT** be collapsed, merged, or selectively omitted outside their declared profile. Their independence ensures that each representational dimension remains structurally stable, while their interdependence guarantees that the artifact contains sufficient fidelity to be interpreted consistently in future computational environments.

3.3 Separation of Representation vs. Computation

Representation is defined solely by the EDM schema; persistence is defined solely by the .ddna envelope; and computation is strictly external to this architecture. This boundary prevents representational drift by ensuring that no computational operator—whether retrieval, ranking, appraisal, synthesis, or weighting—can modify representational structure or embed model-specific assumptions.

The architecture **STRICTLY PROHIBITS**:

- The embedding of computed features or latent-space descriptors within representational fields.
- The storage of model-internal activations.
- The definition of fields whose representational semantics depend on a specific model.

This separation is a normative constraint; any system reading an EDM artifact **MUST** apply governance and validation before computation and **MUST NOT** alter representational content. The architecture thereby ensures multi-vendor interoperability and long-range interpretive stability.

3.4 Identity, Sovereignty & Longevity Principles

EDM defines affective artifacts as sovereign, rights-bearing objects. Sovereignty is enforced through in-band governance metadata—including jurisdiction, retention, and rights descriptors—that remains structurally attached to the artifact throughout its lifecycle. Identity association, where required, is mediated through a neutral binding mechanism that preserves the strict separation of representation from identity. **VitaPass** is the reference implementation of this mechanism; implementations may use alternative binding systems that satisfy the neutrality requirements defined in Section 6.2.

This separation is the architectural enabler for the **Stateless Configuration**: because identity is defined as a detachable binding rather than an intrinsic representational property, systems may generate, process, and discard fully populated affective artifacts without creating persistent identity associations. Sovereignty is architecturally enforced by embedding governance rules directly within the artifact and strictly prohibiting downstream systems from detaching, overriding, or reinterpreting this metadata. Version identifiers serve as immutable anchors for long-range interpretability, ensuring that artifacts remain intelligible across decades regardless of changes in model architecture or computational paradigms.

3.5 Architecture Topology

The architecture is defined by three primary visual schemas that illustrate the structural relationships between components:

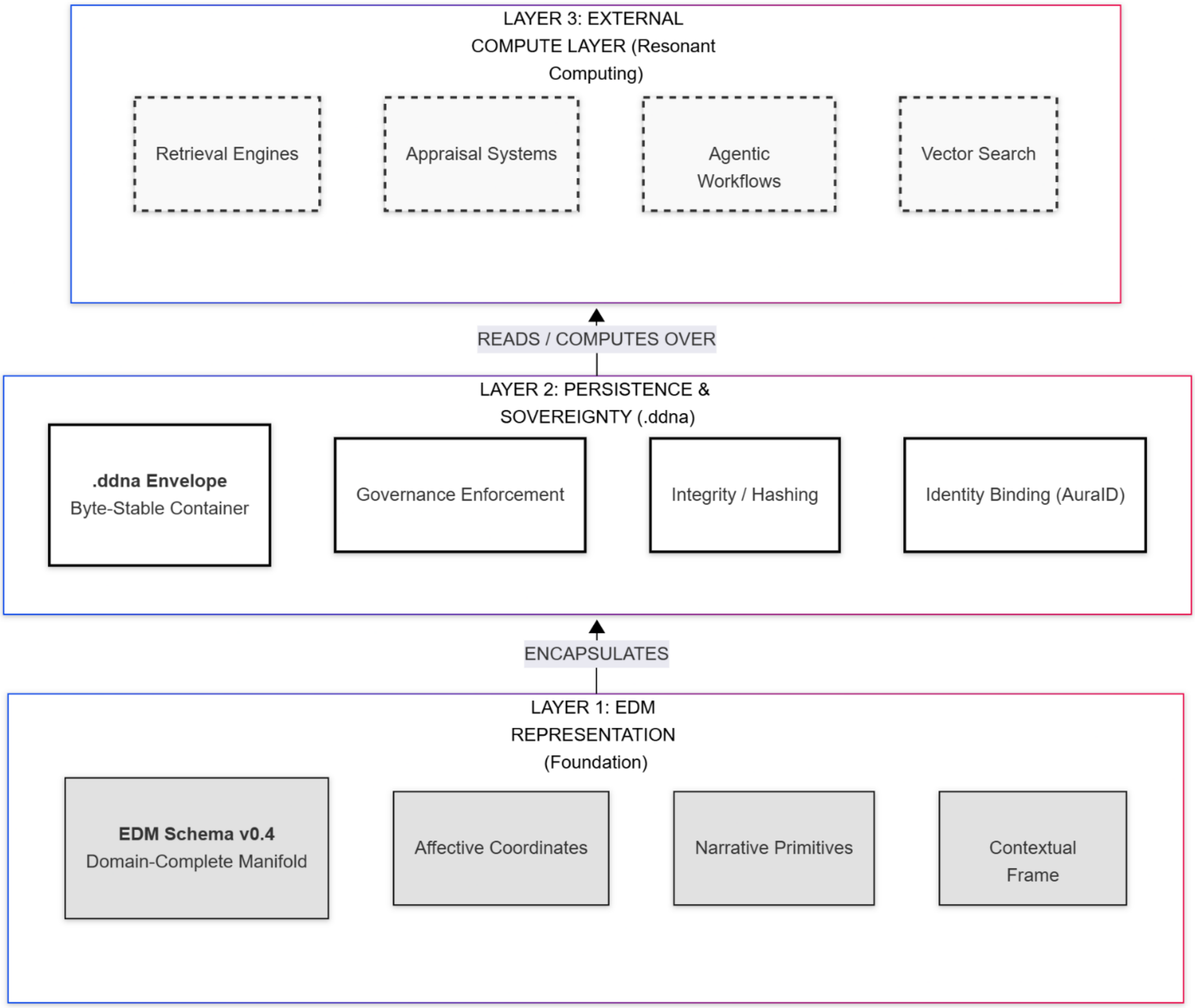

Figure 1: *The Stack Topology: Illustrates the strict layering of the system, positioning the EDM Representation Layer as the foundational plane, the .ddna Envelope Layer as the encapsulation boundary, and the External Compute Layer as the consumer.*

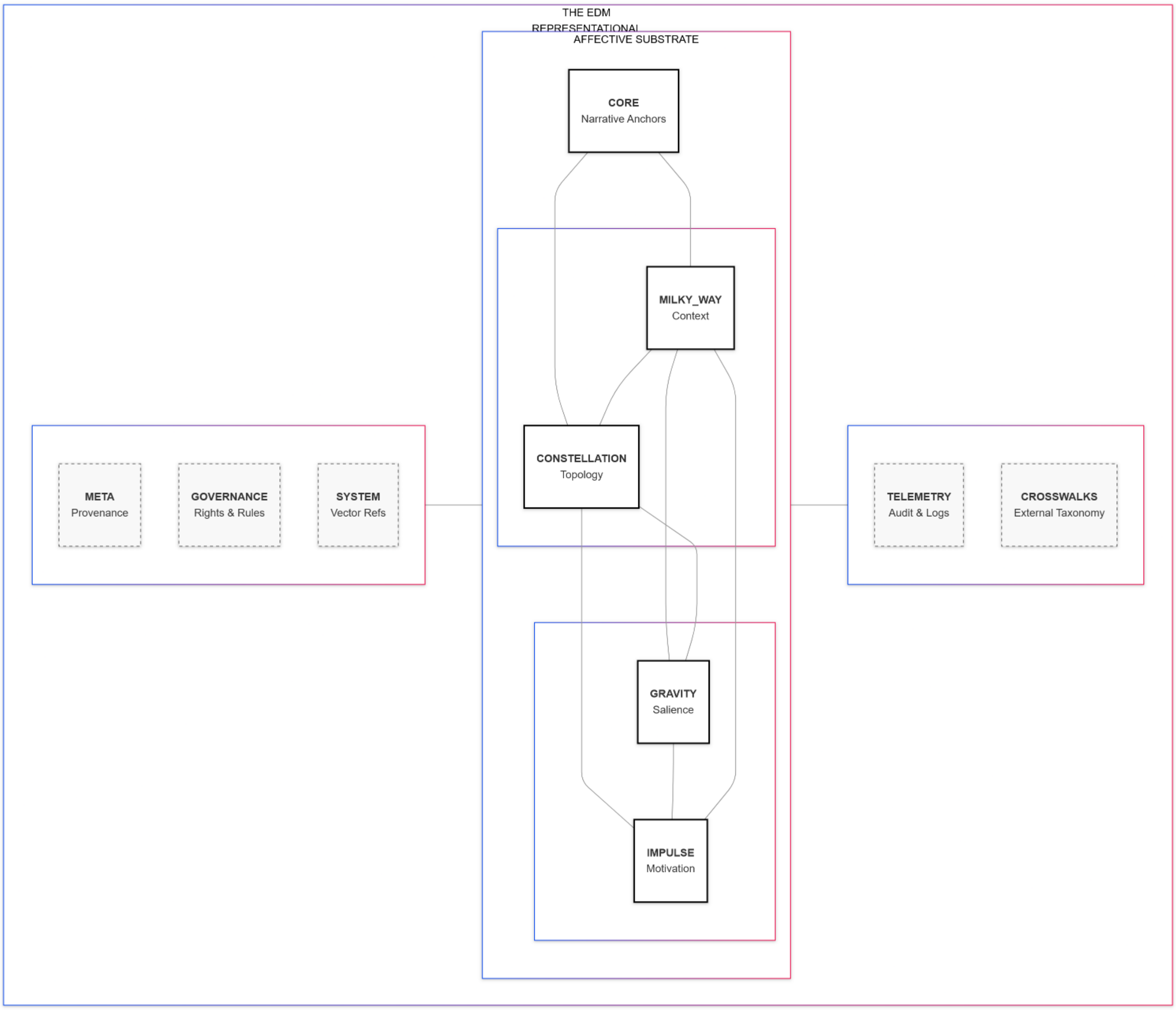

Figure 1: The Representational Manifold Topology: Visualises the interlocking relationship of the ten mandatory domains, distinguishing the central affective geometry (Essential, Constellation, Gravity, Impulse, Milky_Way) from the peripheral governance and metadata structures (Meta, Governance, Telemetry, System, Crosswalks

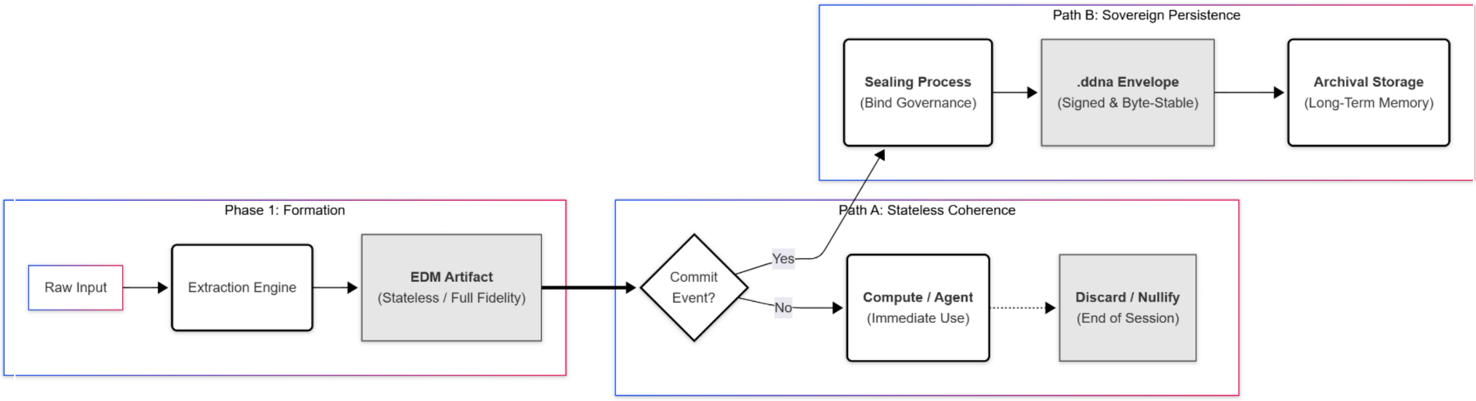

Figure 1*: The Artifact Lifecycle: Demonstrates the unidirectional progression of data from Extraction → Sealing → Transport → Verification → Consumption, highlighting the specific points where governance is enforced, and integrity is verified*

3.6 Operational Modalities

To support flexible integration while enforcing sovereignty, the architecture defines two operational modalities.

Stateless Configuration. A system generates a profile-complete EDM artifact populated according to the schema. In this state, the artifact acts as a transient representational object, enabling immediate affective coherence and context shaping within a session or chat environment. Compliant systems MUST treat these unsealed artifacts as volatile. If the artifact is not committed to a persistent state, sensitive domains—specifically `Milky_Way` and `Gravity` **MUST** be definitively nulled or discarded upon session completion. Narrative and affective state domains **MAY** be retained to enable high-fidelity coherence and provide a structured, anonymised signal for system optimisation.

Persistent Configuration. Upon a commitment event (e.g., user confirmation or system archive command), the artifact is sealed inside a **.ddna envelope**. This binds identity, salience, affective context, and governance metadata to the payload, transforming the transient signal into a sovereign, immutable memory object.

3.7 Implementation Profiles

3.7.1 Overview

EDM defines three implementation profiles that determine the representational depth of a conforming artifact. Each **Profile** specifies a distinct set of domains and fields appropriate to a target deployment context, enabling graduated adoption across the ecosystem without compromising the governance and sovereignty guarantees that apply at every level.

Profiles are orthogonal to conformance levels (Section 3.8). A profile defines *what* is represented — the structural depth of the affective artifact. A conformance level defines *how* that artifact is governed and verified. The two dimensions compose independently, allowing implementations to select the representational depth appropriate to their use case while independently choosing the trust posture required by their operational context.

3.7.2 Profile Declarations

The `meta.profile` field **MUST** be present. Canonical values are `essential`, `extended`, and `full`. An artifact extracted under a registered partner profile (Section 3.7.6) **MUST** declare its partner profile ID as the `meta.profile` value using the `partner:` prefix (e.g., `partner:com.deepadata.journaling.v1`). The `partner:` prefix is the discriminant — a reader encountering a `partner:`-prefixed value knows immediately that registry resolution is required to determine the full field manifest, without maintaining or checking an enum of canonical values. A reader encountering a `meta.profile` value that is neither a canonical profile name nor `partner:`-prefixed **MUST** treat the artifact as non-conforming. Omission of this field renders the artifact non-conforming to EDM v0.8.0 and later.

3.7.3 Essential Profile

The **Essential Profile** defines the minimum viable representational artifact for session emotional coherence and real-time retrieval. It establishes the narrative foundation of an experience through the Core domain, encodes primary affective topology through a minimal Constellation subset, and attaches the governance and provenance metadata required for rights-bearing operation. This profile is designed for memory platforms, agent frameworks, and AI assistants that require structured affective context without therapeutic or longitudinal depth.

The Essential Profile activates five domains: `Meta`, `Core`, `Constellation`, `Governance`, and `Telemetry`. Domains outside this set **MUST** be omitted. The exact fields required within each domain are defined in Section 4.

3.7.4 Extended Profile

The **Extended Profile** extends the Essential Profile with full affective topology, contextual grounding, and salience geometry. It activates the complete `Constellation` domain, providing the full emotional and narrative coordinate set, and introduces the `Milky_Way` and `Gravity` domains to situate the experience within specific environmental and relational coordinates and encode its salience dynamics. This profile is designed for journaling applications, companion AI with longitudinal memory, and workplace wellness platforms where relational and temporal depth are required.

The Extended Profile activates eight domains: all Essential Profile domains plus `Milky_Way`, `Gravity`, and `Impulse`. Domains outside this set **MUST** be omitted. The exact fields required within each domain are defined in Section 4.

3.7.5 Full Profile

The **Full Profile** activates all ten domains of the EDM representational manifold, providing complete affective, contextual, salience, intent, and governance geometry. It is required for regulated contexts, enterprise compliance, and certification-eligible artifacts where complete representational depth is necessary for audit and rights enforcement. The Extended and Full Profiles are both eligible for Certified conformance (Section 3.8). The Essential Profile is not eligible for Certified conformance.

All ten domains **MUST** be fully populated. All fields **MUST** be populated from the source content. A field **MUST** be set to null only where the source provides no expressed basis for extraction.

3.7.6 Partner Profiles

A **Partner Profile** is a named, versioned field manifest registered against a canonical base profile, enabling vertical-specific field selection while preserving the Profile Completeness invariant. Partner profiles are defined and maintained by platform operators and registered through a conforming partner profile registry.

Partner profiles address a structural requirement that canonical profiles cannot satisfy: a deployment context may require a specific subset of fields from one or more domains, optimised for token efficiency, storage cost, and retrieval precision, without accepting the full domain set of the canonical base profile. A therapy platform requires session continuity fields and clinical `Impulse` depth but not contextual `Milky_Way` fields. A **significance wiki** requires `gravity` retrieval handles and `arc_type` but not the full `Core` emotional primitive set. Partner profiles resolve this mismatch structurally, at registration time, without violating the architecture's representational invariants.

The cost model operates across two independent levers. The **base profile** determines LLM inference cost: extraction runs to the full field set of the declared base profile. The **included_fields manifest** determines API response payload: only declared fields are returned to the caller. If all required fields are available within the Extended profile, declaring Extended as the base profile avoids the inference cost of Full-only fields. If even one Full-only field is required, the base profile **MUST** be Full, and Full inference cost applies regardless of how few fields are declared in `included_fields`. The partner profile designer **SHOULD** select the shallowest canonical base profile that contains all required fields.

3.7.6.1 Partner Profile Structure

A conforming partner profile registration **MUST** declare the following properties:

`profile_id` — A globally unique identifier in reverse-DNS notation (e.g., `com.deepadata.journaling.v1`). **MUST** be lowercase and dot separated. The full `meta.profile` value for artifacts produced under this profile is `partner:<profile_id>` (e.g., `partner:com.deepadata.journaling.v1`). The profile identifier **MUST** be immutable after registration.

`base_profile` — The canonical profile from which this partner profile draws its field set. **MUST** be one of `essential`, `extended`, or `full`. Extraction runs to the full field set of the declared base profile. The base profile is the extraction target; the partner profile is the response filter.

`included_fields` — An ordered array of field path strings declaring the fields to be returned in the API response (e.g., `constellation.arc_type`, `gravity.emotional_weight`). All declared fields **MUST** be present in the base profile's field set. Fields not declared in `included_fields` are extracted and stored server-side, remaining available for re-projection to any partner profile or canonical base profile without re-extraction.

`edm_version` — The EDM schema version this partner profile is built on (e.g., `0.8.3`). Partner profiles referencing deprecated fields are marked invalid at the next EDM minor version increment in which those fields are deprecated.

`extensions_schema` — Optional. A namespace and field declaration for partner-specific extension fields. **MUST** follow the extensions architecture defined in Section 3.9. Extension fields declared in a registered partner profile become queryable via arc_signatures at the API key scope only. Extension field weights **MUST NOT** cross API key boundaries.

3.7.6.2 Profile Completeness Invariant Preservation

The Profile Completeness invariant (Section 3.7.7) **MUST** be satisfied at the level of the declared base profile. Extraction runs to the full domain and field set of the base profile. The partner profile filters the API response; it does not constrain extraction or storage. The complete base-profile artifact is always stored server-side, ensuring that governance, auditability, and sovereignty guarantees apply to the full extracted artifact. The stored artifact can be re-projected to a different partner profile or to the canonical base profile at any future time without re-extraction.

3.7.6.3 meta.profile Declaration

An artifact extracted under a partner profile **MUST** set `meta.profile` to `partner:` followed by the registered profile identifier (e.g., `partner:com.deepadata.journaling.v1`). Bare profile identifiers without the prefix are non-conforming. The `meta.profile` value is immutable after extraction per the Profile Immutability invariant (Section 3.7.7).

A conforming reader encountering a `partner:`-prefixed `meta.profile` value **MUST**: (1) identify the value as a partner profile artifact by the presence of the `partner:` prefix, without an enum check or registry access; (2) attempt registry resolution to obtain the base profile and field manifest; (3) if resolution succeeds, validate canonical fields against the resolved base profile schema restricted to `included_fields`; (4) if resolution fails, treat declared canonical fields as fully interpretable and **MUST NOT** attempt completeness validation. Partner profile artifacts carry a conditional portability constraint: offline structural validity is limited to canonical field semantics. Canonical profile artifacts (`essential`, `extended`, `full`) carry no such constraint — they are fully self-describing and do not require registry access for validation.

3.7.6.4 Interoperability Rules

Canonical fields present in a partner profile artifact are fully interoperable. Any EDM v0.8.3-conforming reader can interpret canonical field values without knowledge of the specific partner profile. Extension fields declared in a partner profile’s `extensions_schema` require bilateral schema agreement. A reader without the partner’s extensions schema **MUST** ignore extension fields gracefully. VitaPass subject binding and domain portability classification (Section 3.9) are unaffected by partner profile declarations. Portable domains remain portable regardless of which fields are declared in `included_fields`. System and Extensions domains remain vendor-local.

3.7.6.5 Certification Eligibility

A partner profile artifact is eligible for Certified conformance (Section 3.8.4) if it satisfies the **Certification Minimum Bar**: the artifact **MUST** contain at minimum ten populated affective fields. The affective field set comprises all fields within the Core, Constellation, Milky_Way, Gravity, and Impulse domains. Meta, Governance, Telemetry, System, Crosswalks, and Extensions fields do not count toward the threshold. A conforming certification authority enforces this minimum bar at partner profile registration time. The standard defines the bar; it does not mandate a specific certification authority.

3.7.6.6 Reference Partner Profiles

Four reference partner profiles are defined for EDM v0.8.0. These profiles are normative reference implementations. Full field manifests are defined in the schema package at `schemas/partner-profiles/` in the edm-spec repository. A representative example artifact is provided in Appendix B (Artifact B4).

`partner:com.deepadata.journaling.v1` — Base: Extended. 16 fields (14 affective). Focus: identity continuity, emotional weight, arc pattern over time. Certification eligible.

`partner:com.deepadata.therapy.v1` — Base: Full. 24 fields (24 affective). Focus: session continuity, clinical depth, recurrence, tethering, and therapeutic arc. Certification eligible.

`partner:com.deepadata.companion.v1` — Base: Extended. 18 fields (18 affective). Focus: relational depth, emotional state tracking, identity continuity across sessions. Certification eligible.

`partner:com.deepadata.wiki.v1` — Base: Extended. 17 fields (17 affective). Focus: significance classification for personal knowledge management and wiki article generation. `core.narrative` is intentionally excluded; the wiki generator operates on raw source text. Certification eligible.

3.7.6.7 Schema Version Independence

Partner profiles are versioned independently of EDM schema versions via the `edm_version` property. EDM schema version increments do not invalidate partner profiles unless the increment deprecates fields referenced in `included_fields`. When a referenced field is deprecated, the partner profile **MUST** be updated before the next registration renewal. A partner profile referencing deprecated fields is marked invalid by a conforming registry and **MUST NOT** be used as the declared profile for new artifacts.

3.7.7 Profile Invariants

The following invariants apply to all profiles and constitute normative architectural constraints:

Profile Completeness: An artifact **MUST** contain only the domains defined for its declared profile. Domains not defined for the declared profile **MUST** be omitted. Inclusion of undeclared domains renders the artifact non-conforming.

Exact Field Set: An artifact **MUST** contain only the fields defined for its declared profile. Fields not defined for the declared profile **MUST** be omitted. Omission of required fields is prohibited.

Null Representation: A field **MUST** be set to null only where the source content provides no expressed basis for extraction.

Profile Immutability: The `meta.profile` value **MUST** be set at extraction time and **MUST NOT** be modified thereafter. An artifact's profile is fixed for its lifetime.

Profile Portability: Cross-vendor subject binding through the .ddna envelope is available at all three profiles. Portability eligibility is independent of profile depth and is defined by the sovereignty architecture in Section 7.

3.8 Conformance Levels

3.8.1 Overview

EDM defines three conformance levels that specify how an artifact is governed and verified. **Conformance levels** enable graduated trust: development environments may operate at Compliant level, production platforms may seal artifacts for tamper-evidence, and regulated contexts may require third-party certification.

Conformance levels are orthogonal to implementation profiles (Section 3.7). A conformance level defines *how* an artifact is governed — the trust and verification posture applied to the representational artifact. A profile defines *what* is extracted — the depth of affective representation. The two dimensions compose independently, allowing implementations to select the trust posture appropriate to their operational context without constraining their representational depth.

3.8.2 Level 1 – Compliant

A **Compliant artifact** is a valid EDM JSON object that conforms to the declared profile schema. Governance metadata is self-declared by the producing system. No cryptographic signing is required.

Eligible Profiles: Essential, Extended, Full, Partner.

The artifact **MUST** validate against the EDM schema for the declared profile. All profile-required fields **MUST** be populated per Section 3.7. Governance completeness requirements are defined in Section 6.

3.8.3 Level 2 – Sealed

A **Sealed artifact** is encapsulated in a .ddna envelope, providing cryptographic tamper-evidence and ensuring governance metadata travels immutably with the artifact across system and vendor boundaries.

Eligible Profiles: Essential, Extended, Full, Partner.

All Level 1 (Compliant) requirements **MUST** be satisfied. The artifact **MUST** be encapsulated in a valid .ddna envelope. Envelope structure and cryptographic proof requirements are defined in Section 7.

Stateless TTL Enforcement: Unsealed artifacts containing populated `milky_way` or `gravity` domains **MUST** be nulled or sealed within 24 hours of creation. This ceiling is normative; compliant implementations **MAY** enforce a shorter window.

3.8.4 Level 3 – Certified

A **Certified artifact** is a Sealed artifact that has been countersigned by a **certification authority**. The certification proof attests to schema conformance, governance completeness, consent basis validity, and jurisdiction-appropriate field handling.

Eligible Profiles: Extended, Full, and any Partner Profile (Section 3.7.6) whose `included_fields` manifest contains ten or more affective fields (the Certification Minimum Bar). The Essential profile contains nine affective fields and is not certification-eligible. This is a design boundary: Essential is the minimum viable profile for agent and memory platform contexts where complete affective depth is not required. Certification requires demonstrable affective depth that the Essential profile does not assert.

All Level 2 (Sealed) requirements **MUST** be satisfied. The artifact **MUST** satisfy the Certification Minimum Bar. Certification proof requirements and approved certification endpoints are defined in Section 7.

3.8.5 Profile x Conformance Matrix

The following matrix defines valid profile and conformance level combinations. Combinations not listed are non-conforming.

Profile	Compliant	Sealed	Certified
Essential	✓	✓	—
Extended	✓	✓	✓
Full	✓	✓	✓
Partner	✓	✓	✓ if ≥10 affective fields

Certified conformance is available to Extended, Full, and Partner Profile artifacts meeting the Certification Minimum Bar (10+ affective fields, Section 3.7.6.5). Essential Profile artifacts are not eligible for Certified conformance. All profile and conformance combinations are valid at Compliant and Sealed levels.

3.9 Domain Portability Classification

EDM domains are classified into two portability categories that define their behaviour during cross-system artifact transfer. This classification is a normative architectural constraint; implementing systems **MUST** honour these rules when processing artifacts under VitaPass or any cross-vendor portability mechanism.

Portable Domains (9). The following domains constitute the canonical artifact and **MUST** be preserved in full during any cross-system transfer: `Meta`, `Core`, `Constellation`, `Milky_Way`, `Gravity`, `Impulse`, `Governance`, `Telemetry`, and `Crosswalks`. These domains represent the subject's affective context, provenance, and rights — data that belongs to the subject and travels with the artifact regardless of originating platform. Receiving systems **MUST** accept and preserve all portable domains without modification.

Vendor-Local Domains (2). The `system` and `extensions` domains are classified as vendor-local. System domain fields (`embeddings`, `vector_refs`, `waypoint_ids`) are pointers into the originating platform's infrastructure and have no meaning outside that environment. Receiving systems **MUST** generate new `system` domain content from the portable canonical record rather than attempting to interpret originating `system` fields. The `extensions` domain contains partner-namespaced enrichments representing the originating partner's proprietary semantic layer and intellectual property. Extensions **MUST NOT** be assumed portable across system boundaries; receiving systems **MUST** treat `extensions` content as opaque unless an explicit bilateral agreement governs its interpretation. The seal hash records whether `extensions` were present at seal time through an `extensions_hash`, enabling proof-of-existence verification without content disclosure.

This classification is independent of the identity binding mechanism used. Whether portability is mediated through VitaPass, an alternative sovereign identity system, or a direct bilateral transfer, the portability classification of each domain remains fixed. The portability architecture does not govern user data access rights under applicable law; those rights are defined by the Governance domain and the applicable regulatory framework.

4. EDM v0.8.3 Schema Architecture

The EDM v0.8.3 schema defines the representational substrate through which affective context is encoded as a structured, governed, model-agnostic data object. EDM treats affective context as a **structured manifold** where each domain contributes a distinct coordinate dimension essential for representational completeness. The schema enforces profile completeness (Section 3.7.7), byte-stable structure, explicit nullability, and a strict separation between representational semantics and computational processes.

For the complete normative definition of fields, data types, enumerations, and validation constraints, refer to **Appendix A: Canonical EDM Schema**.

4.1 Formal Representational Geometry

To formalise the distinction between latent inference and explicit structure, EDM defines affective representation as a composite state:

$$\mathbf{R}(\mathbf{x}) = \big(\mathbf{\Sigma}(\mathbf{x}), \mathbf{E}(\mathbf{x})\big)$$

In this formulation, $\mathbf{\Sigma}(\mathbf{x})$ denotes the semantic substrate—encompassing the lexical, multimodal, or embedding-based content of the input—while $\mathbf{E}(\mathbf{x})$ denotes the structured affective mapping encoded via the EDM schema. Unlike contemporary architectures where affective state is treated as a transient derivative of $\mathbf{\Sigma}(\mathbf{x})$, this protocol instantiates $\mathbf{E}(\mathbf{x})$ as an explicit, independent data object. The representational manifold defined by $\mathbf{E}(\mathbf{x})$ comprises up to ten domains, with the active domain set determined by the declared implementation profile (Section 3.7). The Full Profile establishes a fixed coordinate system across all ten domains for narrative, emotional, and motivational state.

This geometry imposes a strict computational boundary. Downstream systems operate only on the fully realised represented state, denoted as:

$$\mathbf{I}(\mathbf{x}) = \mathbf{C}\big(\mathbf{R}(\mathbf{x})\big),$$

where $\mathbf{C}$ represents the external computational operator. This ordering enforces the architectural invariant that **representation must precede computation**. No computational system is permitted to modify, reinterpret, or derive new representational semantics for any domain within $\mathbf{E}(\mathbf{x})$ after artifact creation. By mathematically decoupling the affective structure $\mathbf{E}(\mathbf{x})$ from the inference engine $(\mathbf{C})$, the architecture ensures that emotional context remains a stable, governable invariant across heterogeneous model generations.

4.2 Representation Invariants

The EDM schema defines non-negotiable representational invariants that ensure each artifact remains structurally complete and geometrically stable. **Profile Completeness** requires that an artifact contain only the domains and fields defined for its declared profile, with absence expressed only through explicit null values. For the full definition of profile-conditional domain sets see Section 3.7. The **No Omission**, **No Inference** rule mandates that extractors populate fields only with observable or directly expressed attributes; no field may contain content derived from psychological diagnosis, behavioural prediction, or latent reconstruction. **Canonical Enumerations** must be used for all categorical fields, and **Byte-Stable Payload Rules** prohibit reordering, normalization, or structural modification during handling. Finally, **Interpretive Stability** ensures that the artifact remains semantically interpretable across model generations by avoiding model-specific latent geometry.

4.3 The Affective Manifold (Domains)

The primary function of EDM is to stabilise emotional meaning into five distinct representational planes. These domains constitute the central representational manifold, encoding the qualitative and quantitative dimensions of experience without reliance on transient inference.

Narrative Primitives (`core`)

The `core` domain establishes the structural foundation of the experience, anchoring the context through positional primitives. It identifies the `anchor` as the central referent and the `spark` as the initiating trigger, distinguishing the event from its emotional consequence. The domain maps the `wound`—the point of disruption or vulnerability—and the `fuel` that sustains the emotional energy, enabling systems to structure the causal geometry of an experience. By encoding the `bridge` as the transitional structure and the `echo` as the residual continuity, the schema provides deterministic coordinates for identifying narrative progression.

Emotional Topology (`Constellation`)

The Constellation domain maps the qualitative landscape of the affect. It captures the emotion_primary and emotion_subtone using canonical vocabularies, providing a stable categorical reference that persists across model versions. It records the subject's archetypal identity via the narrative_archetype — which of the twelve canonical identity archetypes the subject embodies, an identity rather than a structural role. The domain distinguishes between routine processing and fundamental shifts in state through the transformational_pivot marker, and captures explicit realisations via expressed_insight. To link text with somatic state, it records the somatic_signature, ensuring that the quality and certainty of the signal are represented as explicitly as its content.

Contextual Framing (`Milky_Way`)

The `Milky_Way` domain situates the emotion within specific environmental and relational coordinates. It defines the `location_context`, `associated_people`, and `event_type`, providing the environmental data necessary to differentiate context without containing the emotional signal itself. This separation ensures that context remains distinct from affect, allowing systems to compute situational relevance independently of emotional intensity.

Salience & Dynamics (`Gravity`)

The Gravity domain quantifies the salience dynamics of the memory. It encodes `emotional_weight` as a scalar value, defining the artifact's resistance to decay and its priority within retrieval operations. It tracks the `recurrence_pattern` to distinguish between isolated events and chronic themes, and defines the `tether_type` to identify the mechanism binding the emotion to the subject. This domain provides the quantitative inputs required for resonance-weighted retrieval, enabling systems to prioritise memory recall based on their structural significance rather than recency or lexicon alone.

Motivational Orientation (`Impulse`)

The Impulse domain encodes the directional vector of the experience. It describes the `drive_state` and `urgency` of the representation, structuring motivational energy as a vector rather than a static label. By explicitly defining the directionality, the protocol allows downstream agents to align their responses with the user's directional intent, ensuring that the system supports the user's movement toward or away from a stimulus without inferring moral judgment.

4.4 Governance & Sovereignty Architecture

EDM differentiates itself from standard metadata schemas by treating governance as a structural invariant equal in importance to the content itself. The `Governance` domain establishes the regulatory framework governing artifact handling. Unlike `Meta` (which records artifact identity and provenance), `Governance` defines the legal obligations, retention requirements, subject rights, and compliance conditions that systems **MUST** enforce. This separation ensures that regulatory constraints remain distinct from extraction metadata while maintaining in-band sovereignty guarantees. It defines the **jurisdiction** governing the data, the `retention_policy` determining its lifespan, and the `subject_rights` (such as portability and erasure) that travel with the object. This in-band governance model ensures that compliance and user rights are preserved even as the artifact moves across vendor boundaries.

4.5 Infrastructure & Interoperability

The remaining domains provide the technical scaffolding required for integration, provenance, and cross-system translation. The `System` domain manages vector compatibility through stable `vector_refs`, ensuring the artifact remains independent of any specific embedding model. The `Telemetry` domain records static extraction metadata, including `entry_confidence` and the source model identifier, ensuring auditability of the extraction process without embedding dynamic usage statistics. The `Meta` domain anchors the artifact in time and provenance, while the `Crosswalks` domain ensures long-range interoperability by providing non-destructive mappings to external taxonomies—allowing the protocol to evolve alongside affective science without destabilising its core representation.

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5. Structural Invariants & Normative Guarantees

The EDM v0.8.3 schema is governed by a set of structural invariants that define the representational boundaries of the architecture. These invariants regulate how artifacts are formed, preserved, and interpreted across systems and temporal horizons. They apply exclusively to the EDM representational layer and precede any persistence guarantees introduced by the .ddna envelope. Their purpose is to maintain the stability of the affective manifold defined in Section 4, ensuring that representational semantics remain uniform across implementations and model generations. The profile-level invariants introduced in Section 3.7.7 are expanded normatively here. These invariants constitute the non-negotiable architectural rules of the Emotional Data Architecture Stack; they specify constraints on structure, semantics, and population that **MUST** be satisfied for an artifact to be considered valid.

5.1 Profile Completeness

Profile completeness requires that an artifact contain only the domains and fields defined for its declared implementation profile. Because each profile defines a specific representational depth appropriate to its deployment context, the inclusion of undeclared domains or the omission of required domains violates structural integrity and interrupts interpretive continuity. The declared profile determines the exact domain set; no domain outside that set may be present, and no required domain may be absent.

For the Full Profile, all ten domains of the EDM manifold **MUST** be present. For the Essential and Extended profiles, only the domains defined in Section 3.7 **MUST** be present. Domain sets for each profile are normatively defined in Section 3.7.3, 3.7.4, and 3.7.5 respectively.

Profile completeness is the prerequisite for schema stability, long-term preservation, and cross-model interoperability. It supersedes the domain completeness model of earlier EDM versions, in which all ten domains were mandatory in every artifact regardless of deployment context.

5.2 Population Invariants: No Omission, No Inference

The population of the schema is governed by two complementary rules. First, the No Omission rule mandates that all fields defined for the declared profile **MUST** be present. Fields within the declared profile that lack extractable content **MUST** be populated with explicit null values; silent omission is prohibited. Fields outside the declared profile **MUST** be omitted entirely — null population of out-of-profile fields is not permitted. Every representational absence within the profile must be encoded as a structural fact. Second, the No Inference rule dictates that extractors may populate fields only with observable or directly expressed attributes. No field may contain content derived from psychological diagnosis, behavioural prediction, or latent reconstruction. Population **MUST** be derived from computational extraction or direct declaration, ensuring the artifact records observed structure, interpreted or explicit, rather than inferred mental states. Together, these rules ensure that EDM artifacts reflect the representational substrate of the source material without introducing inference drift or latent behavioural semantics.

5.3 Canonical Enumerations

Enumerated fields, including affect classes, subtones, relational stance identifiers, structural markers, consent bases, and jurisdiction tags **MUST** use the canonical enumerations defined by the v0.8.3 schema. Enumerations **MUST NOT** be extended, reinterpreted, or replaced by vendor-specific alternatives. Any modification to these sets requires an explicit schema version change under the evolution principles defined in Section 11. Canonical enumerations ensure that implementations across heterogeneous platforms reference the same representational semantics, maintaining alignment with the formal affective manifold and preventing semantic fragmentation.

5.4 Byte-Stable Payload Rules

Byte stability requires that the serialized form of an EDM artifact remain identical across storage systems, transport mechanisms, and platform boundaries. Artifacts **MUST NOT** be rewritten, normalized, or structurally modified during handling.

This invariant **STRICTLY PROHIBITS** the reordering of fields, automated renaming, whitespace or formatting normalization, the embedding of vector data in representation fields, and any transport-layer conversions that alter structure. Byte-stable encoding ensures that integrity checks, governance enforcement, schema validation, and long-term archival practices can rely on deterministic structural identity.

5.5 Interpretive Stability Across Model Generations

Interpretive stability requires that an EDM artifact remain semantically interpretable across successive generations of models, regardless of changes in architecture, inference mechanisms, or embedding spaces. This invariant is enabled by the schema's architecture-neutral design: **representational semantics** are defined externally to model behaviour. To achieve this stability, the architecture mandates consistent domain meanings, explicit field definitions, the avoidance of model-specific latent geometry, and strict independence from inference pathways. Interpretive stability positions the EDM manifold as a long-term representational substrate capable of surviving computational and architectural change without semantic loss.

5.6 Temporal Coherence & Provenance Anchoring

Temporal coherence requires that temporal metadata—including timestamps, extraction context, and lifecycle markers—remain intact and unaltered across all systems that store or process an artifact. Provenance anchoring ensures that the temporal signature of an artifact remains attached to the representational payload without reinterpretation. This invariant supports auditability, chronological reconstruction, multi-model lineage tracing, and deterministic reasoning about extraction context. Temporal coherence ensures that time-dependent relationships across artifacts are maintained without requiring model-internal reconstruction or historical inference.

5.7 Schema Evolution Principles

Schema evolution follows controlled, versioned change. All modifications **MUST** adhere to strict architectural principles: backward-compatible additions require a minor version increment, while structural changes or modifications to representational semantics require a major version increment. Deprecations **MUST** be explicit and accompanied by defined downgrade procedures. Forward compatibility requires that future implementations interpret legacy artifacts without representational loss. Cross-version interpretability **MUST** be preserved through stable field meanings and deterministic migration pathways, preventing fragmentation of the representational ecosystem and maintaining coherence as the schema evolves.

6. Identity, Sovereignty & Governance Architecture

Identity, sovereignty, and governance constitute structural properties encoded directly within the EDM schema and preserved by the .ddna envelope. These properties define the representational conditions under which an EDM artifact remains intact, rights-bearing, and interpretable across storage, execution, or vendor environments. They do not describe operational behaviour or enforcement mechanisms; rather, they establish the immutable constraints that downstream systems **MUST** honour. Any modification to governance descriptors or identity-binding metadata constitutes a formal representational update and **MUST** follow schema-governed processes.

6.1 Sovereign Artifact Model

The EDM artifact operates as a sovereign representational object. Its content, governance descriptors, and envelope-level integrity markers form a single, byte-stable structure. Governance fields—including consent basis, jurisdiction, retention semantics, visibility rules, exportability constraints, and subject-rights descriptors—reside within the Governance domain and remain structurally attached to the artifact for its entire lifecycle. Sovereignty is an architectural condition; governance **MUST NOT** be detached, overridden, externalised, or reinterpreted without altering the representational object itself. The .ddna envelope enforces this condition by preserving governance metadata exactly as encoded and ensuring that all header–payload coupling remains byte-stable. Consequently, sovereignty persists across system boundaries as a structural invariant of the representational object.

6.2 VitaPass Identity Binding

VitaPass provides an optional identity-binding mechanism designed specifically for affective context; a portable, human-centric address that maintains strict separation between subject metadata and emotional representation. Unlike general-purpose identifiers such as UUIDs, which provide uniqueness without semantic protection, VitaPass is architected for the specific sensitivity of emotional data. It exists as a distinct layer precisely because affective artifacts carry a quality of human experience that demands a higher standard of sovereignty than conventional identity resolution provides.

Identity-binding metadata resides exclusively within designated metadata fields and **MUST NOT** appear within narrative, affective, contextual, or structural domains. The presence of a VitaPass binding does not modify the meaning of representational fields. This separation is the architectural enabler for the

Stateless Configuration: because identity is defined as a detachable binding rather than an intrinsic representational property, systems may generate, process, and discard fully populated affective artifacts without creating persistent identity associations.

The meta.`owner_user_id` field accepts VitaPass, UUID, or vendor-specific identifiers. Only VitaPass provides portable cross-vendor subject binding with the sovereignty guarantees appropriate to emotional data. Implementations using alternative identifier types accept reduced portability and reduced subject protection as an architectural consequence.

The VitaPass consent, verification, and disclosure infrastructure is defined in a separate specification and operates above the representational layer defined by this architecture.

6.3 Retention, Consent & Jurisdiction Descriptors

Retention windows, consent bases, jurisdiction codes, and visibility classes are fixed, declarative fields within the Governance domain. Their **governance semantics** are defined exclusively by the schema and remain independent of organisational policy or implementation detail. These descriptors specify the **representational conditions** under which an artifact must be stored, processed, or transferred. Because these fields form part of the representational object, they remain authoritative throughout the artifact's lifetime. Changes to these descriptors constitute representational updates governed by schema version discipline. Schema-extrinsic reinterpretation, custom enumerations, or context-dependent semantics are **STRICTLY PROHIBITED** to preserve interoperability and long-range consistency.

6.4 Exportability Controls

Exportability is encoded as a representational constraint that specifies the permissible scope of artifact relocation. These descriptors **DEFINE** whether transfer is structurally allowed, which destination classes or jurisdictions are authorized, and how governance conditions persist across boundaries. The .ddna envelope **MUST** preserve exportability descriptors exactly as encoded. These descriptors do not implement enforcement mechanisms themselves; rather, they define the representational constraints that downstream systems **MUST** evaluate prior to any transfer operation. Exportability functions solely as a structural boundary condition within the artifact.

6.5 Rights Descriptors & Subject-Aligned Controls

Subject-rights descriptors—including erasure, revocation, visibility modification, and access rights—are declarative structures within the Governance domain. They do not describe enforcement workflows or behavioural procedures; they

ESTABLISH the representational obligations that **MUST** be honoured prior to processing the artifact. These descriptors apply uniformly to the representational payload and all associated identity-binding metadata. They **MUST NOT** introduce narrative, affective, or contextual semantics. Their sole function is to ensure subject alignment through structural embedding.

6.6 Multi-Party Context Structures

EDM supports multiple subject associations within a single representational object. Multi-party structures consist of independent identity-binding metadata and corresponding governance descriptors. These structures indicate only the existence of multiple subject associations and **MUST NOT** be used to encode relational, interpersonal, or behavioural meaning. Differential jurisdiction, visibility, or rights descriptors across subjects **MUST NOT** alter representational semantics. The blending, aggregation, or inference of shared attributes is **STRICTLY PROHIBITED**. Multi-party context remains a structural configuration of independently governed associations.

6.7 Reduced-Governance Configuration (Stateless Mode)

The schema defines a reduced-governance configuration for contexts where identity association, extended retention, or subject-level governance is unnecessary. Unsealed artifacts containing populated `milky_way` or `gravity` domains **MUST** be nulled or sealed within 24 hours of creation. This ceiling is normative; compliant implementations **MAY** enforce a shorter window. The TTL enforcement requirement is defined in Section 3.8.3. Narrative and affective state domains (`core`, `constellation`, `impulse`) **MAY** be retained to enable high-fidelity coherence and provide a structured, anonymised signal for system optimisation. This configuration ensures that representational geometry remains identical while eliminating persistent identity footprints, allowing artifacts to function as safe, interoperable training signals for generalised affective alignment. Statelessness is a variation in governance and content scope, not a different representational mode, and remains interoperable with full-governance artifacts through shared schema boundaries.

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7. The .ddna Envelope Architecture

The .ddna envelope provides the persistence and sovereignty layer for the Emotional Data Architecture Stack. While EDM defines the representational manifold, the envelope defines the structural, cryptographic, and governance guarantees required to preserve that representation across storage regimes, computational systems, and temporal horizons. The envelope functions strictly as a container, not a transformation layer; it **MUST NOT** compress, optimise, reinterpret, or reorder representational content. Its sole architectural function is to ensure that an EDM artifact remains intact, verifiable, and governed as a byte-stable object. The envelope constitutes a deterministic, versioned structure whose semantics are fixed by this specification, guaranteeing that representational content, governance descriptors, and integrity material remain co-located and authoritative across heterogeneous environments.

7.1 Purpose of .ddna

The .ddna envelope establishes four normative architectural guarantees. **Persistence** ensures that representational content remains stable across storage systems, migrations, and long-horizon archival. **Sovereignty** mandates that governance descriptors remain structurally inseparable from the payload and cannot be externalised or superseded by platform-level policy. **Integrity** ensures the detectability of any mutation through deterministic cryptographic verification. **Portability** guarantees that the artifact remains interpretable across model families, embedding standards, and vendor ecosystems. These guarantees position the envelope as the authoritative vessel for long-lived affective representation, distinguishing it from transient wire formats.

7.1.1 File Extension and MIME Type Conventions

The .ddna envelope uses standardised file extension and MIME type conventions to ensure consistent identification and handling across systems. Two artifact types are defined:

Artifact Type	Extension	MIME Type	Description
Unsealed EDM artifact	`.edm.json`	`application/json`	Raw EDM artifact for interchange
Sealed .ddna envelope	`.ddna`	`application/vnd.deepadata.ddna+json`	Governed, signed envelope

The `.ddna` extension signals that the file is a governed artifact requiring compliant tooling for proper handling. This convention distinguishes sealed envelopes from unsealed artifacts, enables downstream systems to identify governance requirements, and supports consistent handler association across toolchains. Internal storage systems **MAY** use different conventions, but interchange formats **MUST** use these canonical extensions and MIME types.

7.2 Envelope Structure

The envelope comprises three mandatory components: a **Header** (`ddna_header`), an **EDM Payload** (`edm_payload`), and a **Proof Block** (`proof`).

The Header defines the metadata required for deterministic decoding. It records the envelope version, EDM schema version, identity-binding configuration, governance mode (full or stateless), and lifecycle audit chain.

The Payload stores the unmodified EDM artifact. The envelope **MUST NOT** reformat domain structure, suppress null values, reorder keys, embed vectors, or introduce interpretive transformations. Payload content is preserved byte-for-byte to ensure that the signed hash matches the representational state exactly.

The Proof Block contains the **W3C Data Integrity** proof that cryptographically binds the header and payload. The proof enables verification that the envelope content has not been modified since signing. Integrity validation is a mandatory prerequisite to any read or write interaction with the payload.

7.3 Cryptographic Guarantees

Envelope integrity is enforced through W3C Data Integrity Proofs using the `eddsa-jcs-2022` cryptosuite or a declared equivalent. This approach provides cryptographic binding of the header and payload without requiring separate hash fields, ensuring that mutation, drift, or tampering cannot occur without detection.

All compliant systems **MUST** validate the proof before processing or reading the payload. The verification method is declared within the proof block and **MUST** be resolvable by any compliant verifier. This architecture maintains an unambiguous chain of custody for sensitive emotional data across all system and vendor boundaries.

Implementation guidance for the signing and verification process, including canonicalization, hash computation, signature generation, and encoding conventions, is defined in the companion document `DDNA_SIGNING_MODEL.md`.

7.4 Migration and Version Anchoring

Each envelope declares explicit version identifiers for both the envelope format and the enclosed EDM schema. Version anchoring ensures interpretability across schema evolution. Schema migration follows deterministic rules: minor schema changes **MUST** remain backward-compatible; structural changes require major version increments; and deprecated fields **MUST** remain legible through defined downgrade semantics. Older artifacts must continue to be interpretable by newer readers without re-serialisation. No system is permitted to reinterpret a payload under a schema version other than the one declared in the header.

7.5 Governance Embedding

Governance descriptors are encoded in-band and constitute a structural component of the artifact's identity. These descriptors — including consent basis, jurisdictional constraints, retention windows, visibility rules, exportability conditions, and subject-rights declarations — **MUST** remain intact throughout the artifact's lifecycle. Governance metadata is authoritative; it **MUST NOT** be detached, weakened, or overridden by external policy engines or storage systems. Integrity rules apply uniformly to both governance and representational content, ensuring that subject-aligned rights persist across all system boundaries and temporal states.

7.6 Transport and Storage Guarantees

The envelope remains invariant under transport, replication, and storage. Deterministic serialisation guarantees that the artifact is encoded identically across platforms and toolchains. Payload content **MUST NOT** be optimised, reordered, compressed, or transformed in any manner that alters its canonical byte sequence. Governance descriptors **MUST NOT** be separated from the payload. Integrity guarantees **MUST** remain enforceable across distributed systems, archival infrastructures, and cross-jurisdictional movement, ensuring byte-identical persistence and unambiguous interpretability across the artifact's full lifecycle.

7.7 Envelope Serialisation Rules

Serialisation is governed by canonical ordering per **RFC 8785 JSON Canonicalization Scheme (JCS)**, the explicit representation of null values, and stable formatting rules. The envelope **MUST NOT** permit optional suppression of fields, platform-dependent variation, or re-encoding of representational content. Deterministic serialisation ensures that any compliant system produces identical byte sequences for a given artifact, enabling cryptographic verification, long-horizon archival, and cross-vendor interoperability without reliance on proprietary decoding logic. The normative serialisation requirements of JCS are defined in RFC 8785.

7.8 Stateless Envelope Configuration

The architecture supports a stateless envelope configuration for environments requiring secure transport without long-term persistence or identity binding. In this configuration the .ddna envelope maintains integrity and provenance guarantees while enforcing the privacy constraints of the stateless modality.

Within the payload of a stateless envelope, identity-binding metadata **MUST** be omitted. Domains not defined for the declared profile **MUST** be omitted in accordance with the Profile Completeness invariant (Section 3.7.7). Where the declared profile includes `milky_way` or `gravity` domains and the artifact is not sealed, those domains **MUST** have all fields set to null within 24 hours of creation. Sealing the artifact within the TTL window is the alternative path that preserves those domains under full governance. Governance descriptors are reduced to the minimal structural fields required for immediate processing.

The TTL ceiling governing unsealed stateless artifacts is defined in Section 3.8.3. The schema-layer governance constraints for stateless mode are defined in Section 6.7. This section defines the envelope-layer structural requirements only.

This ensures that representational semantics remain unchanged and interoperable while strictly preventing the persistence of sensitive context or identity footprints.

7.9 Long-Range Interpretability

The envelope is designed to remain interpretable across extended temporal horizons. This is achieved through immutable representational semantics, deterministic serialisation, cryptographic verification, explicit version anchoring, and in-band governance persistence. The envelope **does not** define computational behaviour; its role is to preserve representational content as a stable, governed, and verifiable object independent of the inference systems operating above it. This ensures that EDM artifacts remain intelligible across evolving model architectures, embedding geometries, and regulatory regimes.

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8. Future Directions & Evolution Path

The Emotional Data Model (EDM) and the .ddna envelope define a representational and persistence architecture designed to remain stable across computational, regulatory, and archival environments. Evolution of the architecture **MUST** preserve established semantics, maintain domain geometry, and uphold all governance invariants. Future revisions **MUST** operate strictly within the representational boundary defined by the current schema and **MUST NOT** introduce inference pathways, modify existing field meaning, or alter the structural role of any domain. This section defines the representational constraints that govern forward evolution and ensure that artifacts created under v0.8.3 remain valid and structurally interpretable under later versions.

8.1 Versioning as a Structural Commitment

Version identifiers form an intrinsic part of the representational contract and determine the precise schema under which an artifact must be interpreted. Evolution of EDM **MUST** remain strictly backward-interpretable: artifacts produced under earlier versions **MUST** remain structurally legible without reconstruction, approximation, or reinterpreted logic. Any additions to the schema **MUST** occur only within defined extension boundaries and **MUST NOT** alter existing domain geometry or field semantics. Deprecation **MUST NOT** remove interpretability, and no revision may modify the meaning of a previously defined representational construct. This discipline ensures that representational continuity is preserved across schema generations.

8.2 Controlled Expansion of Representational Scope

Expansion of representational scope is permissible only when new structures demonstrate stable, schema-level semantics, non-interference with existing domains, and compatibility with cross-domain geometry. Initial incorporation **MUST** occur through crosswalk references, allowing external constructs to align with EDM fields without altering internal representational semantics. Promotion to optional schema structures is permissible only when definitional stability is established and **MUST NOT** introduce behavioural attributes, inferred semantics, or model-dependent constructs. Representation **MUST** remain independent of interpretive frameworks, and expansion **MUST NOT** affect the structural meaning of existing fields.

8.3 Multimodal Extension Boundaries

Future incorporation of multimodal descriptors **MUST** follow the same stability path applied to text-derived fields. Multimodal extensions **SHOULD** initially be introduced through crosswalk references or external schema extensions to validate representational utility. Promotion to normative schema substructures is permissible only when definitional stability is established, and the constructs demonstrate non-interference with existing domains. Any multimodal field **MUST** encode explicit, non-inferential descriptors and **MUST NOT** depend on temporal, behavioural, or sensory-derived interpretation. Extensions **MUST** preserve domain geometry, maintain schema neutrality, and operate without imposing latent or model-specific constructs.

8.4 Structural Research Trajectories

The architecture externalises affective context as a governed, schema-defined structure, establishing a stable substrate for research independent of model internals. Any analysis performed using EDM artifacts **MUST** remain within the representational boundary and **MUST NOT** depend on model activations, latent geometries, or inference mechanisms. Structural comparison, schema-conformance assessment, and representational variance studies **MAY** operate only on artifact-level fields as encoded. These activities do not influence schema evolution and **MUST NOT** modify representational semantics. EDM's evolution remains governed solely by schema discipline, version rules, and domain-geometry invariants, decoupling structural research from the volatility of inference engines.

8.5 Cross-Ecosystem Interoperability

EDM and .ddna maintain interoperability across heterogeneous computational and archival environments through deterministic serialisation, explicit version anchoring, canonical null handling, and immutable governance embedding. Interoperability arises from structural invariants rather than behavioural compatibility. Systems that store, transport, or encapsulate EDM artifacts **MUST** preserve envelope integrity and **MUST NOT** alter representational content, governance descriptors, or domain geometry. The representational contract remains authoritative across all environments in which the artifact is present, ensuring that affective context retains its structural identity regardless of the vendor or platform.

8.6 Alignment with Global Standards

EDM evolution **MUST** remain compatible with external standards governing structured data, provenance, archival persistence, and governance metadata. Compatibility **MUST NOT** modify EDM's representational semantics or domain boundaries. Serialisation rules, governance invariants, and representational geometry **MUST** remain canonical and independent of any external interpretive frameworks. Alignment **MAY** occur only at the boundary layer, through crosswalks or metadata associations that do not alter or reinterpret internal schema structures, ensuring that the protocol can co-exist with broader ecosystem standards without compromising its core definitions.

8.7 Summary

Evolution of EDM is governed by strict representational discipline. Version identifiers anchor interpretive semantics across generations; extension boundaries restrict permissible additions; multimodal constructs **MUST** remain explicit and non-inferential; structural analysis operates strictly on artifact-level fields; and interoperability is preserved through deterministic serialisation and in-band governance invariants. These constraints ensure that EDM artifacts remain stable, interpretable, and structurally coherent across future computational eras. Expansion is permissible only when it preserves the representational manifold and maintains the permanence of governance and domain semantics.

9. Security Architecture

The security architecture defines the structural guarantees required to preserve the integrity, sovereignty, and interpretability of EDM v0.8.3 artifacts and .ddna v1.1 envelopes across storage, transport, and multi-vendor computational environments. Because EDM externalises affective context as explicit, schema-governed representation, the security model **MUST** prevent mutation, reinterpretation, omission, or detachment of any representational or governance component. The architecture does not prescribe operational security frameworks, cryptographic algorithms, or platform-specific implementations; its scope is limited to the representational, structural, and validation requirements that all compliant systems **MUST** satisfy to ensure long-range integrity. Security in the Emotional Data Architecture Stack is achieved through deterministic serialisation, in-band governance semantics, mandatory profile completeness, immutable envelope coupling, and validation procedures that **MUST** precede any read, access, or computation. These guarantees ensure that EDM artifacts remain authoritative, version-anchored, and interpretable irrespective of platform behaviour, vendor infrastructure, or inference mechanisms.

9.1 Data Minimisation & Stateless Hygiene

EDM enforces strict representational boundaries to minimise data exposure. Fields within the declared profile that lack extractable content **MUST** appear as explicit nulls; fields outside the declared profile **MUST** be omitted entirely. Omission-based signalling within the declared profile is prohibited. Unknown or absent values **MUST** appear as explicit nulls in accordance with schema semantics; no omission-based signalling is permitted. To further minimise risk, systems operating in the **Stateless Configuration MUST** definitively nullify sensitive domains—specifically `Milky_Way` (Context) and `Gravity` (Salience)—before any persistence event, ensuring that unsealed artifacts do not retain identifiable or longitudinal data. This minimisation constraint ensures that EDM artifacts remain schema-complete, rights-aligned, and structurally consistent across heterogeneous environments.

9.2 Structural Security (In-Band Enforcement)

Security is enforced at the representational level rather than the transport level. Governance descriptors—including consent basis, retention policy, jurisdiction codes, visibility constraints, exportability rules, and subject-rights declarations—are embedded directly within the representational object and **MUST** remain attached to the artifact throughout its lifecycle. No system may detach, override, reinterpret, or substitute governance metadata. All systems **MUST** validate governance descriptors before accessing or processing the EDM payload. Because governance is encoded in-band, representational constraints are enforced independent of platform security features, ensuring consistent treatment across heterogeneous or untrusted infrastructures.

9.3 Threat Model

The architectural threat model identifies classes of violations that would compromise structural, representational, or governance integrity. These include the unauthorised modification of schema fields, removal or alteration of governance metadata, schema drift arising from non-canonical serialisation, mutation of null semantics, reordering of keys, introduction of platform-specific identifiers, and unauthorised rewriting during storage or transport. EDM mitigates these risks through canonical serialisation, explicit null preservation, profile completeness, immutable envelope binding, and version-anchored interpretive semantics. While the protocol addresses structural integrity, implementation-specific vectors—including **side-channel attacks** and **timing attacks** during cryptographic verification—remain the responsibility of the runtime environment and must be mitigated by the implementation layer. Any violation that alters structural identity **MUST** be detectable by integrity verification processes.

9.4 At-Rest Requirements

Artifacts stored at rest **MUST** retain byte-stable identity. Storage systems **MUST NOT** rewrite serialised content, apply transformations that alter canonical ordering, compress or reinterpret structure, or modify governance descriptors. Representational fields **MUST** remain exactly as emitted. Storage-layer metadata **MUST** remain external to the representational object and **MUST NOT** be introduced into EDM domains. These requirements ensure that artifacts remain structurally identical to their original emission and retain full interpretive fidelity over time.

9.5 In-Transit Requirements

Artifacts in transit **MUST** remain structurally and semantically identical to their stored form. Transport mechanisms **MUST NOT** re-serialise, re-encode, compress, or transform artifacts in ways that alter canonical field ordering, null representation, formatting, or governance metadata. Upon receipt, systems **MUST** validate the .ddna envelope, verify header–payload coupling, perform integrity checks, and enforce governance semantics before accessing representational content. Systems **MUST NOT** process an artifact prior to full validation. These requirements ensure that artifacts cannot be silently mutated or detached during transfer.

9.6 Multi-Vendor and Zero-Trust Considerations

The architecture assumes a zero-trust ecosystem in which no vendor, model family, platform, or inference system is considered authoritative. Consequently, all protection arises from representational constraints and envelope-level invariants rather than from shared trust models or platform guarantees. Canonical serialisation, explicit governance, domain completeness, version-anchored interpretive semantics, and cryptographically verifiable envelope integrity ensure that artifacts remain valid and enforceable across incompatible, decentralised, or adversarial systems.

9.7 Sovereignty Beyond Platform Boundaries

EDM artifacts retain sovereignty across all environments. Governance descriptors—including consent, jurisdiction, retention, rights, and visibility—remain embedded within the representational object and are preserved unchanged within the .ddna envelope. Platforms **MUST NOT** weaken, detach, or reinterpret these descriptors without violating structural validity. Identity binding is anchored through VitaPass and does not rely on platform identifiers or access-control systems. Sovereignty, therefore, persists as an architectural property independent of vendor infrastructure or operational context.

9.8 Tamper Detection and Provenance Integrity

The .ddna envelope ensures provenance integrity through W3C Data Integrity Proofs. The proof block cryptographically binds both header and payload, with the `created` timestamp providing temporal anchoring. Version identifiers define the interpretive frame under which the artifact must be read. Audit markers provide structural traceability without embedding behavioural logs or runtime events. Any compliant system **MUST** verify these integrity markers before accessing representational content. These guarantees ensure that artifacts remain authentic, unmodified, and semantically aligned with the normative specification defined by EDM v0.8.3.

10. Interoperability & Crosswalks

Interoperability in EDM is defined as the capacity for external taxonomies and classification systems to reference EDM artifacts without modifying, reshaping, or influencing representational semantics. Because EDM operates as a closed, schema-governed architecture, compatibility with heterogeneous ecosystems **MUST** occur exclusively at the boundary layer through a strictly referential structure. The `Crosswalks` domain provides this boundary mechanism, enabling external systems to attach deterministic identifiers to specific EDM fields while ensuring that the artifact itself remains sovereign, byte-stable, and semantically unchanged. Interoperability thus operates outside the representational manifold and **MUST NOT** introduce additional semantics, behavioural assumptions, interpretive structures, or theoretical models.

10.1 Purpose and Design Philosophy

The `crosswalk` mechanism supports structural alignment between EDM artifacts and diverse external frameworks—ranging from established psychological taxonomies to **emerging sensing modalities** such as vocal prosody, facial landmarks, and empathetic inference signals. Its purpose is strictly limited to establishing machine-readable correspondence between EDM's representational fields and external categorical systems. `Crosswalks` **MUST NOT** alter, enrich, constrain, or reinterpret EDM domains; they serve solely as deterministic references for alignment. This design is governed by three architectural principles: **Non-Destructive Alignment** ensures that all EDM fields and structural definitions remain unchanged regardless of external referencing; **Schema Preservation** mandates that the EDM schema remains authoritative across all environments; and **Isolation of External Semantics** requires that crosswalks do not import interpretive definitions, intensity geometries, or behavioural logic. `Crosswalks` provide stable identifiers that reference EDM fields without participating in **representational meaning**.

10.2 Mandatory Structure

The Crosswalks domain is a **REQUIRED** top-level domain in Full Profile artifacts. Essential and Extended Profile artifacts **MUST NOT** include the Crosswalks domain per the Profile Completeness invariant defined in Section 3.7.7. Each `crosswalk` entry **MUST** contain a canonical identifier for the external taxonomy, a stable external label or code, the explicit EDM field or domain it references, and the version identifier for the external taxonomy. Descriptive text, commentary, or interpretive metadata is **STRICTLY PROHIBITED**. All entries must remain semantically inert with respect to EDM and **MUST NOT** affect the representational manifold. The presence of the `crosswalk` domain does not modify representational meaning, and its absence does not remove representational content; its function is strictly referential, ensuring structural consistency regardless of external taxonomic associations.

10.3 Non-Destructive Mapping

All `crosswalk` behaviour is governed by strict non-destructive guarantees. `Crosswalks` **MAY** append new reference entries, but they **MUST NOT** modify, redefine, or reinterpret any EDM domain or field. The representational semantics of EDM are immutable and **MUST NOT** be influenced by external mapping. No crosswalk entry may introduce inferred structure or conceptual meaning. The removal of the crosswalk domain **MUST** revert the artifact to a byte-identical form. These constraints ensure that `crosswalks` operate exclusively at the interoperability boundary and **cannot** influence representation, persistence, governance, or schema evolution.

10.4 External Taxonomy Linkage

EDM v0.8.3 supports linkage to external taxonomies through referential identifiers. This linkage is structural and non-semantic. External systems—such as categorical affect schemes, structured intensity identifiers, contextual classifications, and international metadata standards—**MAY** be referenced through stable codes, but their interpretive logic **MUST NOT** be imported into EDM. These references **DO NOT** introduce new representational primitives, modify existing representational semantics, or establish theoretical dependencies. `Crosswalks,` therefore, provide only compatibility alignment and **DO NOT** participate in representational geometry.

10.5 Future-Proofing and Extensibility

The interoperability architecture is designed for long-term stability across evolving taxonomic ecosystems. Each `crosswalk` entry **MUST** contain an explicit version identifier for the external taxonomy to prevent ambiguity or reinterpretation as external systems evolve. Changes in external frameworks **MUST NOT** trigger any modification to EDM's domains, fields, or enumerations. `Crosswalk` updates are isolated operations that **MUST** remain additive and non-invasive. New taxonomies **MAY** be referenced without altering any representational content. Governance descriptors, rights metadata, jurisdictional fields, and sovereignty invariants **MUST** remain unaffected by `crosswalk` behaviour. These constraints ensure that EDM artifacts maintain stable representational semantics across future analytic, regulatory, or interoperability landscapes while supporting deterministic referencing of external classification systems.

11. Version Evolution & Migration Discipline

Version evolution within the Emotional Data Architecture Stack is governed by strict representational discipline. Because EDM encodes affective context as an explicit, schema-bound object rather than a model-internal construct, any change to the schema must preserve the interpretive integrity of all existing artifacts. Versioning operates as an architectural mechanism rather than an implementation detail; it defines how representational semantics remain stable across revisions, how new fields may be introduced without disturbing prior meaning, and how systems interpret artifacts over extended temporal horizons. The .ddna envelope reinforces these guarantees by binding the schema version identifier within the header, ensuring that decoding requirements remain deterministic across all environments. Schema evolution is intentionally conservative; while versions may extend representational capacity or refine constraints, they **MUST NOT** reinterpret or repurpose existing fields. The primary goal of the versioning architecture is to guarantee that artifacts created under earlier schema versions remain legible and semantically stable across future environments, vendors, and computational ecosystems.

11.1 Version Lineage (v0.1 → v0.2 → v0.3 → v0.4.0→ v0.5.0 → v0.6.0 → v0.7.0 → v0.8.0 → v0.8.1 → v0.8.2 → v0.8.3)

The EDM version lineage reflects the incremental expansion of representational structure while maintaining all prior semantics. v0.1 introduced the initial domain layout and core governance constructs. v0.2 stabilised domain boundaries, normalised narrative primitives, and formalised governance invariants. v0.3 introduced the crosswalk architecture, refined governance fields, and established alignment with the .ddna envelope layer. v0.4.0 solidified these definitions into a normative standard. v0.5.0 updated the envelope cryptographic model to W3C Data Integrity Proofs and expanded five canonical enumerations — emotion_primary, relational_dynamics, drive_state, coping_style, and adaptation_trajectory to capture expressive states identified through systematic multi-provider evaluation. v0.6.0 introduces Implementation Profiles (Essential, Extended, Full), Conformance Levels (Compliant, Sealed, Certified), and Profile Completeness as the canonical structural invariant, replacing the prior Domain Completeness model. Nine enumeration values were added across five fields — emotion_primary (shame), relational_dynamics (grandparent_grandchild, friend, couple, colleague), narrative_archetype (orphan), tether_type (identity, self), and motivational_orientation (authenticity) — validated through end-to-end corpus evaluation.

v0.7.0 extends v0.6.0 with seven structural additions and four field deprecations. The `arc_type` field was added to the `Constellation` domain, providing twelve canonical narrative arc values (betrayal, liberation, grief, discovery, resistance, bond, moral_awakening, transformation, reconciliation, reckoning, threshold, exile) with Zod union validation permitting free-text extension. The `extensions` domain was introduced as the eleventh domain of the Full Profile, providing a partner-namespaced open object for post-extraction semantic enrichments; `extensions` are included in the seal hash when present at seal time and are classified as vendor-local (non-portable) under the domain portability architecture formalised in Section 3.9. The `meta.source_timestamp` field was added to anchor source content temporally independent of extraction time. Three enumeration values were added to `emotion_primary` (disappointment, relief, frustration) and two to `narrative_arc` (loss, confrontation), validated through end-to-end corpus evaluation. Hard enum constraints were relaxed on four fields --- `tether_type`, `recurrence_pattern`, `coping_style`, and `relational_dynamics` --- to accept free-text values where no canonical value accurately represents the extracted content, implementing the two-tier canonical-plus-extension enum model. Four fields were deprecated: `gravity.legacy_embed`, `telemetry.alignment_delta`, `system.indices.sector_weights,` and `crosswalks.HMD_v2_memory_type`. Deprecated fields **MUST NOT** appear in v0.8.0 artifacts; their semantics are frozen per Section 11.5. No existing field meanings or enumeration values have been modified or removed.

v0.8.0 extends v0.7.0 with the Partner Profile system and two arc_type additions. Partner Profiles (Section 3.7.6) introduce a registered, versioned mechanism for vertical field selection: a partner profile declares a named subset of fields from a canonical base profile, enabling token-efficient extraction tailored to a specific deployment context while preserving the Profile Completeness invariant at the base profile level. The base profile determines LLM inference cost; the `included_fields` manifest determines API response payload. Four reference partner profiles are defined: journaling, therapy, companion, and wiki. The `meta.profile` field is extended to a two-tier model: canonical values (`essential`, `extended`, `full`) remain valid and unchanged; partner profile IDs are declared using the `partner:` prefix (e.g., `partner:com.deepadata.therapy.v1`), enabling unambiguous discrimination without an enum check. Two arc_type values were added — `gratitude` (moments of thankfulness and appreciation, distinct from the emotion state joy) and `authenticity` (self-alignment and identity congruence) — bringing the canonical arc_type set to fourteen values. Both were validated through WhatMattered benchmark evaluation prior to promotion to canonical status. The Certification Minimum Bar is formalised: certification eligibility requires ten or more populated affective fields from the Core, Constellation, Milky_Way, Gravity, and Impulse domains, opening Certified conformance to partner profiles meeting this threshold. Section 3.7.6 (former Profile Invariants) is renumbered to Section 3.7.7 to accommodate the new

Partner Profiles section. No fields were deprecated in v0.8.0. No existing field meanings or enumeration values have been modified or removed.

v0.8.1 is a references-and-errata patch. Sofroniew et al. (2026) and Memory Bear A-MBER (2026) are added to §14.6 as related work. The Appendix A meta.version constraint is corrected from “0.7.x” to “0.8.x”, and reference schema enum lists are corrected for nullable-enum conformance per §5.2 (24 fields). The Zenodo publication lineage is reunified under concept DOI 10.5281/zenodo.17808652. No fields are introduced, deprecated, or semantically modified.

v0.8.2 is a schema-reconciliation patch that brings the published JSON Schema fragments into alignment with the canonical vocabulary (ADR-0030: the published specification is the source of truth and downstream implementations generate from it). The narrative_archetype enumeration is corrected to the twelve canonical identity archetypes — hero, caregiver, seeker, sage, lover, outlaw, innocent, magician, creator, everyman, jester, ruler. The field encodes the archetypal identity the subject embodies, not a structural role or literary trope: mentor is removed from the fragment enumeration on this ground, and orphan, which earlier documents listed but no shipped v0.8 fragment enumerated, is likewise outside the canonical set. motivational_orientation gains authenticity (self-alignment and identity congruence), aligning the fragment with the canonical vocabulary. The shared meta.profile fragment is reconciled to the two-tier model (canonical enumeration plus the partner: pattern), and a mis-encoded enumeration on meta.tags is removed. A non-validating x-edm-canonical annotation is added to seven two-tier free-text fields, carrying the preferred canonical vocabulary in machine-readable form. The specification is published to npm (edm-spec@0.8.2). No fields are added, removed, or renamed. v0.8.2 shipped without a standalone document; its changes are documented in the v0.8.1-to-v0.8.2 crosswalk and folded into this v0.8.3 document.

v0.8.3 is a truth-only patch release. The Full Profile composite schema gains meta.source_timestamp in its inline meta block, conforming to the canonical meta fragment: the field, added in v0.7.0, had been strict-rejected by the composite's additionalProperties constraint since v0.8.0. The change is strictly widening — every artifact valid under v0.8.2 remains valid, and Full Profile artifacts carrying meta.source_timestamp now validate as the fragment always intended. The Full Profile field count is corrected to 91, the machine count of top-level fields across the ten domains of the shipped composite; the prior figure of 96 was hand arithmetic that never matched a shipped v0.8 schema. The narrative_archetype description is corrected to the identity-archetype definition throughout this document; the enumeration is unchanged at twelve. A one-story release policy is adopted (docs/RELEASE-POLICY.md): every npm release must have its document staged, npm and Zenodo publication occur together, and no certification claim may cite a version whose document is not published. No fields are introduced, deprecated, or semantically modified beyond the composite conformance correction.

11.2 Migration Principles

Migration between schema versions is strictly structural and **MUST NOT** introduce new interpretation. The architecture mandates that migration be **deterministic**, ensuring that a migration process always produces the same output for a given artifact regardless of vendor or implementation. Migration **MUST NOT** reinterpret existing values; fields encoded under earlier schemas retain their original semantics and may not be transformed to simulate newer constructs. New structures **MUST** be additive; when lifting to a higher version, new fields introduced by the updated schema must default to null or their canonical initial values. Crucially, **governance metadata MUST remain unchanged**; all governance fields must remain intact, unmodified, and authoritative across version transitions. The architecture **STRICTLY PROHIBITS** the compression, merging, collapsing, or reinterpretation of representational elements during migration.

11.3 Backwards Compatibility

Backwards compatibility ensures that systems built for earlier schema versions can process artifacts created under newer versions. This is achieved by anchoring all representational semantics to earlier interpretations and by ensuring that new fields do not modify or extend the meaning of pre-existing structures. When older systems encounter unknown fields, they **MUST** be able to omit them without loss of meaning or ambiguity. Backwards compatibility requires explicit null fallback rules, stable semantics for all earlier fields, and deterministic downgrade profiles encoded at the envelope layer. The architecture **prohibits** the introduction of any field whose presence requires reinterpretation by older systems.

11.4 Lifting and Downgrading Rules

Version transitions follow two controlled, purely structural pathways: lifting (upgrade) and downgrading. **Lifting** requires preserving all existing content unchanged while populating new version-specific fields with null or their canonical initial values. The version identifier **MUST** be updated in both the artifact and the .ddna header, but no semantic transformation is permitted. **Downgrading** requires removing fields that do not exist in the target version while strictly preserving the semantics and byte-level content of fields that remain valid. Systems **MUST NOT** collapse, merge, compress, or reinterpret newer structures during a downgrade. Governance invariants **MUST** remain stable across both lifting and downgrading operations. These processes maintain representational invariants and ensure that artifacts remain authoritative and interpretable within their target schema version.

11.5 Deprecated Structures

Deprecation is treated as a semantic freeze. When a field or domain is deprecated, its semantics become fixed, immutable, and frozen; it **MUST NOT** be repurposed, modified, or reassigned to new meaning in any future version. Its presence in existing artifacts **MUST** remain interpretable, and documentation **MUST** record its deprecated status and intended phasing rules. Deprecation **DOES NOT** mandate removal; existing artifacts retain their deprecated fields unchanged, and only future schema versions may omit them once safe downgrade paths exist. This discipline preserves long-range interpretability for artifacts produced under older schemas.

v0.8.0 Deprecated Fields. The following fields are deprecated in v0.8.0 and **MUST NOT** appear in conforming v0.8.0 artifacts. Their semantics are frozen and immutable. Existing artifacts produced under prior versions retain these fields unchanged and remain valid under their declared schema version.

`gravity.legacy_embed` (boolean | null) — Deprecated. Indicated whether the experience contributes to long-range personal identity structure. Replaced by the `identity_thread` field in Constellation and the extensions domain for partner-specific longitudinal markers. Frozen semantics; **MUST** be null or absent in v0.8.0 artifacts.

`telemetry.alignment_delta` (number | null) — Deprecated. Recorded a computed alignment signal between extraction output and schema expectations. Removed as this metric conflated representational structure with extraction quality assessment, violating the strict layer separation principle. Frozen semantics; **MUST** be absent in v0.8.0 artifacts.

`system.indices.sector_weights` (object) — Deprecated. Encoded weight distribution across memory sectors (episodic, semantic, procedural, emotional, reflective). Removed as sector weighting is a computational concern belonging exclusively to the External Compute Layer, not the representational layer. Frozen semantics; **MUST** be absent in v0.8.0 artifacts.

`crosswalks.HMD_v2_memory_type` (string | null) — Deprecated. Provided a crosswalk mapping to HMD v2 memory type taxonomy. Removed following discontinuation of the HMD v2 taxonomy as a maintained external reference. Existing crosswalk mappings to this taxonomy retain frozen semantics in prior-version artifacts. Frozen; **MUST** be absent in v0.8.0 artifacts.

11.6 Forward Compatibility and Future Schema Extensions

Forward compatibility ensures that artifacts created under earlier schema versions remain valid and interpretable under future versions. This requires that schema evolution follows three architectural constraints. First, **Representational Invariants** must remain intact; new fields and domains **MUST NOT** alter the representational semantics of existing structures. Second, **Extensions** must be predictable and non-breaking; new structures **MUST** appear only at defined extension points and **MUST NOT** disrupt domain boundaries. Third, **Crosswalks** must remain isolated; external taxonomy updates **MUST NOT** influence representational semantics or schema evolution. Finally, **Envelope-Bound Versioning** must remain authoritative, with the `.ddna` header providing the grounding for interpreting artifacts across version transitions. Fifth, **Partner Profile Version Independence** must be preserved; partner profiles are versioned independently of EDM schema versions, and EDM minor version increments **MUST NOT** invalidate partner profiles unless the increment explicitly deprecates fields referenced in a partner profile's `included_fields` manifest.

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12. Conclusion: EDM as the Representational Bedrock

EDM v0.8.3 defines affective context as a structured, governed, and portable data object. It establishes the representational conditions under which emotional state can exist independently of any model, vendor, or computational environment, and remain interpretable across systems and time. By externalising affective state into a schema-defined manifold and binding it to a deterministic persistence envelope, the architecture provides a stable substrate where emotional information can be preserved rather than inferred, inspected rather than guessed, and transported rather than recreated.

The protocol’s structural invariants—profile completeness, deterministic serialisation, and in-band governance—ensure that emotional context is durable, sovereign, and resistant to interpretive drift. EDM does not define how systems should *use* this information; it defines how it must be *represented* so that any future computation, reasoning, or alignment approach can rely on a consistent and governed foundation.

EDM stands alone in this role. It does not predict or prescribe the architectures built on top of it; it simply establishes the representational foundation those systems will require.

Explicit emotional state precedes representation; representation precedes computation.

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13. Contact & Collaboration

Current Status: Open Source (MIT License)

For inquiries regarding:

- Research collaboration

- Technical evaluation access
- Implementation partnerships

Contact: jason@emotionaldatamodel.org or contact@emotionaldatamodel.org

Web: emotionaldatamodel.org

Reference implementations available under MIT license:

- **edm-spec**: Canonical JSON Schema specification and examples
  `github.com/emotional-data-model/edm-spec`
- **ddna-tools**: Sealing and verification CLI
  `github.com/emotional-data-model/ddna-tools`
- **ddna-reader**: Read-only artifact inspection
  `github.com/emotional-data-model/ddna-reader`

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14. References

This reference list documents the foundational standards, specifications, and structural frameworks that inform the EDM representational architecture. References are grouped by architectural function.

14.1 Schema, Serialization & Data Architecture

IETF RFC 8259: The JavaScript Object Notation (JSON) Data Interchange Format.

IETF RFC 8785: JSON Canonicalization Scheme (JCS).

W3C: Architecture of the World Wide Web, Volume One.

W3C: Verifiable Credentials Data Model v2.0.

W3C: Data Integrity 1.0.

W3C: Data Integrity EdDSA Cryptosuites v1.0.

ISO 14721: Open Archival Information System (OAIS) Reference Model.

HL7 FHIR: Fast Healthcare Interoperability Resources (Foundation for Schema-First Governance).

14.2 Cryptographic Standards

IETF RFC 8032: Edwards-Curve Digital Signature Algorithm (EdDSA).

W3C CCG: did:key Method v1.0.

Multiformats: Multibase specification.

14.3 Affective Neuroscience & Cognition

Damasio, A. (1994). *Descartes' Error: Emotion, Reason, and the Human Brain*. (The Somatic Marker Hypothesis).

Panksepp, J. (1998). *Affective Neuroscience: The Foundations of Human and Animal Emotions*. (Primary Emotional Systems).

LeDoux, J. (1996). *The Emotional Brain*. (Threat circuitry and emotional salience).

14.4 Foundational Visionaries

Bush, V. (1945). *As We May Think*. (Associative Trails and Memory Augmentation).

Engelbart, D. (1962). *Augmenting Human Intellect: A Conceptual Framework*. (Computation as Cognitive Scaffold).

Berners-Lee, T. (1989). *Information Management: A Proposal*. (Universal Structured Interoperability).

14.5 Ethics, Privacy & Sovereignty

Floridi, L. (2013). *The Ethics of Information*. (Informational Rights and Ontology).

Solove, D. (2008). *Understanding Privacy*. (Conceptual foundations for retention and context).

Zuboff, S. (2019). *The Age of Surveillance Capitalism*. (The asymmetry EDM governs against).

14.6 Technical Lineage & Related Work

Vaswani et al. (2017). *Attention Is All You Need*. (The Transformer Architecture).

Anthropic. (2023). *Constitutional AI: Harmlessness from AI Feedback*. (Normative alignment).

OpenAI. (2023). *GPT-4 System Card*. (Risk mitigation and model behaviour).

Sofroniew, N. J. et al. (2026). *Functional emotions in large language models*. arXiv:2604.07729. (Mechanistic interpretability evidence for functional emotional states; EDM v0.4.0 predates this finding by six months.)

Memory Bear. (2026). *A-MBER: Affective Memory Benchmark for Emotion Recognition*. arXiv:2604.07017. (Structured affective memory outperforms raw-history and retrieved-memory baselines.)

Appendix A — Canonical EDM v0.8.3 Schema

META

`id`

Type: string | null

Description: Unique identifier for the EDM artifact. Required for provenance, linking, auditing, and .ddna envelope binding.

Constraints: UUID v4. Immutable.

`version`

Type: string

Description: Declares which EDM schema this artifact conforms to and enables safe validation and migration.

Constraints: MUST match "0.8.x".

`created_at`

Type: string

Description: Marks when the artifact was extracted. Anchors temporal provenance.

Constraints: ISO-8601 UTC timestamp. Required.

`updated_at`

Type: string | null

Description: Optional timestamp for post-extraction updates. Never replaces created_at.

Constraints: ISO-8601 UTC timestamp.

`locale`

Type: string | null

Description: Linguistic and cultural context of the source content.

Constraints: BCP-47 language tag (e.g., "en-au", "en-us"), lowercase.

`owner_user_id`

Type: string | null

Description: The artifact owner identifier. VitaPass enables cross-vendor identity resolution; vendor-specific IDs limit portability. MUST be null in stateless mode.

Constraints: VitaPass (recommended), vp-prefixed ULID format, or free-text; null for stateless.

`parent_id`

Type: string | null

Description: Links this artifact to its parent in multi-step extractions or threaded sequences.

Constraints: UUID.

`profile`

Type: string

Description: Declares the implementation profile under which this artifact was extracted. Determines which domains and fields are present. Immutable after extraction.

Constraints: Canonical values: `essential`, `extended`, `full`. Partner profile IDs declared using the `partner:` prefix (e.g., `partner:com.deepadata.therapy.v1`). The `partner:` prefix is the discriminant. See Section 3.7.6.

`visibility`

Type: string

Description: Determines who may view or process the artifact.

Constraints: private, shared, public.

`pii_tier`

Type: string

Description: Classification of personal or sensitive content. Controls retention, masking, and export rules.

Constraints: none, low, moderate, high, extreme.

`source_type`

Type: string

Description: The medium from which the artifact was extracted.

Constraints: text, audio, image, video, mixed.

`source_context`

Type: string | null

Description: Optional narrative describing the input scenario.

Constraints: Free-text (e.g., "therapy session", "journaling", "voice memo").

`consent_basis`

Type: string

Description: Legal basis for storing and processing emotional data.

Constraints: consent, contract, legitimate_interest, none.

`consent_scope`

Type: string | null

Description: What the user consented to.

Constraints: Free-text (e.g., "memory storage", "reflection", "therapeutic use").

`consent_revoked_at`

Type: string | null

Description: If populated, the artifact becomes non-retrievable and non-exportable, except minimal audit form.

Constraints: ISO-8601 UTC timestamp.

`tags`

Type: array

Description: Optional user-defined labels for organizing or filtering artifacts.

Constraints: Short tokens, lowercase recommended.

`source_timestamp`

Type: string | null

Description: The timestamp of the source content being extracted, where known. Anchors the artifact temporally to the original experience rather than the extraction event. Distinct from `created_at`, which records the extraction timestamp. Added in v0.7.0.

Constraints: ISO-8601 UTC timestamp. Optional. NULL where source timestamp is unavailable or not applicable.

CORE

`anchor`

Type: string | null

Description: Central person, object, or theme of the experience.

Constraints: 1–5 words (e.g., "grandmother", "dad's toolbox", "childhood home").

`spark`

Type: string | null

Description: What triggered or initiated the emotional response.

Constraints: 1–5 words (e.g., "finding old photos", "first snow", "phone call").

`wound`

Type: string | null

Description: Emotional pain, loss, or vulnerability present in the experience.

Constraints: 1–5 words (e.g., "abandonment", "regret", "distance", "grief").

`fuel`

Type: string | null

Description: What energized or motivated the experience.

Constraints: 1–5 words (e.g., "love", "shared laughter", "curiosity", "hope").

`bridge`

Type: string | null

Description: Connection or shift between past and present understanding.

Constraints: 1–5 words (e.g., "forgiveness", "acceptance", "returning home").

`echo`

Type: string | null

Description: What continues to resonate or recur from the experience.

Constraints: 1–5 words (e.g., "her laugh", "smell of rain", "city lights").

`narrative`

Type: string | null

Description: A compressed, faithful account of the experience as described by the source, preserving emotional sequence, temporal markers, and stated context without interpretation.

Constraints: 3–5 sentences.

CONSTELLATION

`emotion_primary`

Type: string | null

Description: The dominant emotional quality expressed in the experience.

Constraints: joy, sadness, fear, anger, wonder, peace, tenderness, reverence, shame, pride, anxiety, gratitude, longing, hope, disappointment, relief, frustration. Free-text accepted where no canonical value accurately represents the extracted content (v0.8.0 two-tier enum model).

`emotion_subtone`

Type: array

Description: Secondary emotional nuances that add depth to the primary emotion.

Constraints: 0–4 items (e.g., "bittersweet", "grateful", "nostalgic").

`higher_order_emotion`

Type: string | null

Description: Complex or layered emotional state.

Constraints: Free text (e.g., "awe", "bittersweetness", "pride", "moral elevation").

`meta_emotional_state`

Type: string | null

Description: How the person relates to their own emotional state.

Constraints: Free text (e.g., "acceptance", "resistance", "curiosity", "confusion").

`interpersonal_affect`

Type: string | null

Description: Emotional posture in relational context.

Constraints: Free text (e.g., "warmth", "defensiveness", "openness").

`narrative_arc`

Type: string | null

Description: The underlying trajectory or story movement represented by the moment.

Constraints: overcoming, transformation, connection, reflection, closure, loss, confrontation. Free-text accepted where no canonical value accurately represents the extracted content (v0.8.0 two-tier enum model).

`relational_dynamics`

Type: string | null

Description: The dominant relational configuration influencing the experience.

Constraints: parent_child, romantic_partnership, sibling_bond, friendship, companionship, mentorship, reunion, community_ritual, grief, self_reflection, family, professional, therapeutic, service, adversarial, grandparent_grandchild, friend, couple, colleague. Free-text accepted where no canonical value accurately represents the extracted content (v0.8.0).

`temporal_context`

Type: string | null

Description: The life-stage or temporal frame associated with the experience.

Constraints: childhood, early_adulthood, midlife, late_life, recent, future, timeless.

`memory_type`

Type: string | null

Description: The class of memory the experience represents within personal identity.

Constraints: legacy_artifact, fleeting_moment, milestone, reflection, formative_experience.

`media_format`

Type: string | null

Description: The format through which the experience is captured or expressed.

Constraints: photo, video, audio, text, photo_with_story.

`narrative_archetype`

Type: string | null

Description: The archetypal identity the subject embodies — which of the 12 canonical identity archetypes the subject expresses, not a structural role or story function.

Constraints: hero, caregiver, seeker, sage, lover, outlaw, innocent, magician, creator, everyman, jester, ruler.

`symbolic_anchor`

Type: string | null

Description: A symbolic or material object, place, or element that concentrates meaning.

Constraints: Free text (e.g., "rocking chair", "wedding ring", "grandmother's kitchen").

`relational_perspective`

Type: string | null

Description: The perspective through which the experience is narrated or understood.

Constraints: self, partner, family, friends, community, humanity.

`temporal_rhythm`

Type: string | null

Description: The perceived temporal movement or cadence of the emotional event.

Constraints: still, sudden, rising, fading, recurring, spiraling, dragging, suspended, looping, cyclic.

`identity_thread`

Type: string | null

Description: A succinct expression of how the experience connects to personal identity.

Constraints: Short sentence.

`expressed_insight`

Type: string | null

Description: The explicit realization or conclusion stated by the subject.

Constraints: Short sentence. Extracted, not inferred.

`transformational_pivot`

Type: boolean

Description: Marks experiences the subject explicitly identifies as life-changing or trajectory-altering at time of extraction.

Constraints: true or false.

`somatic_signature`

Type: string | null

Description: Bodily or interoceptive sensations explicitly described by the subject as part of the emotional experience.

Constraints: Short phrases extracted from stated physical sensations only; NULL when not mentioned.

`arc_type`

Type: string | null

Description: The dominant narrative arc pattern of the experience. Captures the overarching structural shape of the emotional trajectory. Added in v0.7.0.

Constraints: Canonical values: betrayal, liberation, grief, discovery, resistance, bond, moral_awakening, transformation, reconciliation, reckoning, threshold, exile, gratitude, authenticity. Free-text accepted where no canonical value accurately represents the extracted content. NULL where no arc pattern is discernible. Extended Profile and Full Profile only. gratitude and authenticity added in v0.8.0.

MILKY_WAY

`event_type`

Type: string | null

Description: Type of event or situation.

Constraints: Free text (e.g., "family gathering", "farewell", "birthday", "reunion").

`location_context`

Type: string | null

Description: Place or spatial context of the experience.

Constraints: Free text (e.g., "grandmother's kitchen", "beach", "hospital").

`associated_people`

Type: array

Description: Individuals referenced in or connected to the experience.

Constraints: Names or roles, properly cased (e.g., "Sarah", "my father", "the nurse").

`visibility_context`

Type: string | null

Description: Defines the intended visibility or sharing scope of the moment.

Constraints: private, family_only, shared_publicly.

`tone_shift`

Type: string | null

Description: Directional emotional change during the experience.

Constraints: Free text (e.g., "loss to gratitude", "fear to relief", "joy to sadness").

GRAVITY

`emotional_weight`

Type: number

Description: The felt intensity of the emotional experience.

Constraints: 0.0–1.0.

`emotional_density`

Type: string | null

Description: How concentrated or layered the emotional content is.

Constraints: low, medium, high.

`valence`

Type: string | null

Description: The qualitative direction of the emotional experience.

Constraints: positive, negative, mixed.

`viscosity`

Type: string | null

Description: The perceived "stickiness" or persistence of the emotion.

Constraints: low, medium, high, enduring, fluid.

`gravity_type`

Type: string | null

Description: Category or nature of the emotional pull.

Constraints: Free text (e.g., "symbolic resonance", "relational bond", "identity anchor").

`tether_type`

Type: string | null

Description: The type of element to which the experience is emotionally anchored.

Constraints: person, symbol, event, place, ritual, object, tradition, identity, self. Free-text accepted where no canonical value accurately represents the extracted content (v0.8.0).

`recall_triggers`

Type: array

Description: Sensory or symbolic cues that reactivate the memory.

Constraints: Short phrases, lowercase (e.g., "smell of bread", "old songs", "rain sounds").

`retrieval_keys`

Type: array

Description: Compact hooks for memory retrieval.

Constraints: 3–6 tokens (e.g., "grandmother", "kitchen", "warmth", "tradition").

`nearby_themes`

Type: array

Description: Adjacent themes or motifs connected to the experience.

Constraints: Concepts or emotions (e.g., "family", "tradition", "loss", "continuity").

`legacy_embed` [DEPRECATED in v0.8.0 — do not use in new artifacts]

Type: boolean

Description: Whether the moment contributes to long-term identity or life-story continuity.

Constraints: true or false.

`recurrence_pattern`

Type: string | null

Description: The temporal recurrence structure extracted from the experience's context, framing, and temporal markers.

Constraints: cyclical, isolated, chronic, emerging; NULL when pattern cannot be determined.

`strength_score`

Type: number

Description: The binding strength of the emotion within the broader emotional network.

Constraints: 0.0–1.0.

`temporal_decay`

Type: string | null

Description: How quickly the emotional intensity is expected to diminish over time.

Constraints: fast, moderate, slow.

`resilience_markers`

Type: array

Description: Indicators of stabilizing or constructive emotional processing.

Constraints: 1–3 items (e.g., "acceptance", "optimism", "continuity").

`adaptation_trajectory`

Type: string | null

Description: How the emotional state is evolving across time.

Constraints: improving, stable, declining, integrative, emerging.

IMPULSE

`primary_energy`

Type: string | null

Description: Dominant emotional or motivational energy in the moment.

Constraints: Lowercase (e.g., "curiosity", "fear", "compassion", "longing").

`drive_state`

Type: string | null

Description: The behavioural direction or movement expressed through the emotional response.

Constraints: explore, approach, avoid, repair, persevere, share, confront, protect, process.

`motivational_orientation`

Type: string | null

Description: The foundational motivational domain guiding the individual's internal stance.

Constraints: belonging, safety, mastery, meaning, autonomy, authenticity.

`temporal_focus`

Type: string | null

Description: The temporal direction toward which the motivational energy is oriented.

Constraints: past, present, future.

`directionality`

Type: string | null

Description: Whether the impulse is self-directed, relational, or expansive beyond self.

Constraints: inward, outward, transcendent.

`social_visibility`

Type: string | null

Description: The social scope within which the impulse is expressed.

Constraints: private, relational, collective.

`urgency`

Type: string | null

Description: The intensity or immediacy of the motivational state.

Constraints: calm, elevated, pressing, acute.

`risk_posture`

Type: string | null

Description: The stance toward uncertainty or potential risk embedded in the moment.

Constraints: cautious, balanced, bold.

`agency_level`

Type: string | null

Description: The perceived ability to act or influence the situation.

Constraints: low, medium, high.

`regulation_state`

Type: string | null

Description: The stability or turbulence of emotional self-regulation.

Constraints: regulated, wavering, dysregulated.

`attachment_style`

Type: string | null

Description: The relational attachment pattern influencing the emotional expression.

Constraints: secure, anxious, avoidant, disorganized.

`coping_style`

Type: string | null

Description: The primary strategy used to manage or integrate the emotional experience.

Constraints: reframe_meaning, seek_support, distract, ritualize, confront, detach, process.

GOVERNANCE

`jurisdiction`

Type: string | null

Description: The regulatory regime governing this artifact. Determines applicable rights, obligations, and compliance conditions.

Constraints: GDPR, CCPA, HIPAA, PIPEDA, LGPD, None, Mixed.

`retention_policy.basis`

Type: string | null

Description: The rationale determining how long this artifact may be retained.

Constraints: user_defined, legal, business_need.

`retention_policy.ttl_days`

Type: number | null

Description: Time-to-live in days for the artifact. Defines retention boundaries enforced by engines and storage layers.

Constraints: Non-negative integer or null.

`retention_policy.on_expiry`

Type: string | null

Description: Required action once retention elapses.

Constraints: soft_delete, hard_delete, anonymize.

`subject_rights.portable`

Type: boolean

Description: Whether the artifact is eligible for portability requests under governing law.

Constraints: true or false.

`subject_rights.erasable`

Type: boolean

Description: Whether the artifact must be erasable on user request or under statutory obligations.

Constraints: true or false.

`subject_rights.explainable`

Type: boolean

Description: Whether systems must provide interpretability or explanation when computing with this artifact.

Constraints: true or false.

`exportability`

Type: string

Description: Controls whether the artifact may be exported in .ddna format.

Constraints: allowed, restricted, forbidden.

`k_anonymity.k`

Type: number | null

Description: Minimum k-anonymity grouping requirement for safe aggregation, export, or research use.

Constraints: Integer ≥ 1 or null.

`k_anonymity.groups`

Type: array

Description: Attribute groups that contribute to anonymity constraints.

Constraints: Quasi-identifier group names (e.g., "region", "age_band").

`policy_labels`

Type: array

Description: Classification labels identifying sensitive categories that invoke stricter governance and export limits.

Constraints: sensitive, children, health, biometrics, financial, none.

`masking_rules`

Type: array

Description: Redaction requirements for identity, location, or narrative details during display or export.

Constraints: Mask type specifications.

TELEMETRY

`entry_confidence`

Type: number

Description: Confidence score assigned by the extractor for the accuracy of the EDM representation.

Constraints: 0.0–1.0.

`extraction_model`

Type: string | null

Description: Identifier of the model or engine used for extraction.

Constraints: Free text (e.g., "DeepaTagger-v0.4.0", "GPT-4").

`extraction_notes`

Type: string | null

Description: Optional notes about extraction quality or contextual factors.

Constraints: Free text.

`alignment_delta` [DEPRECATED in v0.7.0 — do not use in new artifacts]

Type: number | null

Description: Measured deviation between the artifact's encoded affective state and the system's response coherence, used to regulate resonance weighting.

Constraints: -1.0 to +1.0; positive indicates over-alignment, negative indicates under-alignment, near-zero indicates equilibrium.

[extraction_chunking_strategy removed from Telemetry in v0.7.0. Chunking strategy is a vendor pipeline concern, not a representational field. If needed, record via the Extensions domain as a partner-namespaced field.]

SYSTEM

`embeddings`

Type: array

Description: Embedding references associated with this artifact. Provides stable pointers without embedding raw data.

Constraints: Each object must include: provider, sector, dim, quantized, vector_ref. No inline vectors permitted.

`indices.waypoint_ids`

Type: array

Description: Identifiers for retrieval or resonance "waypoints" associated with this artifact.

Constraints: Strings only; no null entries.

`indices.sector_weights` [DEPRECATED in v0.7.0 — do not use in new artifacts]

Type: object

Description: Weight distribution across memory sectors used by recall engines.

Constraints: Must include keys: episodic, semantic, procedural, emotional, reflective. Values 0.0–1.0.

EXTENSIONS

Type: object | null

Description: Open object containing partner-namespaced semantic enrichments added post-extraction by integrating systems. Each top-level key is a reverse-DNS namespace identifier. Added in v0.7.0.

Constraints: Keys MUST be reverse-DNS namespace identifiers (e.g., com.limbic, io.mem0). Values MUST be objects. MUST NOT be populated during standard EDM extraction. Included in the seal hash when present. Vendor-local — not portable across system boundaries (Section 3.9). NULL or absent where no partner enrichments have been applied.

CROSSWALKS

`plutchik_primary`

Type: string | null

Description: Mapping hook to Plutchik's primary emotion taxonomy.

Constraints: Free text or null.

`geneva_emotion_wheel`

Type: string | null

Description: Mapping hook to a Geneva Emotion Wheel category.

Constraints: Free text or null.

`DSM5_specifiers`

Type: string | null

Description: Crosswalk reference to DSM-5 affective specifiers. Referential only; not a clinical diagnosis field.

Constraints: Free text or null.

`HMD_v2_memory_type` [DEPRECATED in v0.8.0 — do not use in new artifacts]

Type: string | null

Description: Crosswalk anchor for "Human Memory Dynamics v2" or equivalent memory-type classification frameworks.

Constraints: Free text or null.

`ISO_27557_labels`

Type: string | null

Description: Alignment point for emerging or future ISO standards relating to emotional or affective data classification.

Constraints: Free text or null.

Appendix B — Example EDM v0.8.3 Artifacts

The following artifacts are **ACTUAL** outputs from the same unstructured source input, demonstrating how the three canonical implementation profiles represent the same emotional content at different depths. Artifact B4 demonstrates the partner profile system using the same source, showing field filtering against the therapy partner profile.

Source Input

Just got these photos and what an absolute ball I had on the weekend competing in my physie comps! About 8 years ago I was diagnosed with psoriatic arthritis which affects my joints and I'm on medication that I will need to take forever to keep them mobile. However, it's continuing to move that really keeps it at bay. Not only does physie do this for me, it's also so good for my mental health, something else that was suffering after covid times. As working mums, we tend to put ourselves last and as a single mum for many years, there was no choice but to be last. I believe we all need something in our lives that we can do just for us, and for me, dance has always been it. I adore teaching my @pose_xx kids but I couldn't get through a ballet or open dance class myself anymore, and my days on the stage as a professional dancer are long gone. Getting to still do physie and perform in front of a crowd makes me feel like I'm still in my 20's and getting to do it with other supportive women is really such a gift. Thanks to @jasonpbxx for talking me in to going back to it 4 years ago @thextphysieclub for such a fun and positive class every week and @bxx_physie for giving us ladies a place to stay young and keep active! #alwaysputonashow #dontstopmoving #justlovetodance

Artifact B1 — Essential Profile


```
{
  "meta": {
    "id": "226d1ab2-741e-480a-b656-32d14717f268",
    "version": "0.8.3",
    "profile": "essential",
    "created_at": "2026-03-09T11:44:45.796Z",
    "owner_user_id": " vp-02HZ4GKWP7XTJY9QN7BN",
    "consent_basis": "consent",
    "visibility": "private",
    "pii_tier": "moderate"
  },
  "core": {
    "anchor": "physie competition",
    "spark": "weekend competition photos",
    "wound": "body betrayal through arthritis",
    "fuel": "supportive women community",
    "bridge": "dancing through physical limits",
    "echo": "feeling twenty again",
  },
  "constellation": {
    "emotion_primary": "joy",
    "emotion_subtone": [
      "empowered",
      "grateful",
      "vitality",
      "supported"
    ],
    "narrative_arc": "overcoming"
  },
  "governance": {
    "jurisdiction": null,
    "retention_policy": {
      "basis": "user_defined",
      "ttl_days": null,
      "on_expiry": "soft_delete"
    },
    "subject_rights": {
      "portable": true,
      "erasable": true,
      "explainable": false
    }
  },
  "telemetry": {
    "entry_confidence": 1,
    "extraction_model": "claude-sonnet-4-20250514"
  }
}
```

Artifact B2 — Extended Profile Artifact


```
{
  "meta": {
    "id": "22fe76b6-06fe-419d-b682-340d2abd3b61",
    "version": "0.8.3",
    "profile": "extended",
    "created_at": "2026-03-09T11:44:57.132Z",
    "owner_user_id": " vp-02HZ4GKWP7XTJY9QN7BN",
    "consent_basis": "consent",
    "visibility": "private",
    "pii_tier": "moderate"
  },
  "core": {
    "anchor": "physie competition",
    "spark": "receiving competition photos",
    "wound": "body betrayed by arthritis",
    "fuel": "supportive women community",
    "bridge": "still performing despite limitations",
    "echo": "feeling twenty again",
    "narrative": "Weekend physie competition photos arrived,
capturing the joy of performing despite psoriatic arthritis
diagnosis eight years ago. The supportive community of women and
weekly classes keep both joints mobile and spirits high. Dancing in
front of crowds resurrects the professional dancer within, defying
physical limitations. What began as Jason's encouragement four years
ago has become essential self-care for this working single mum."
  },
  "constellation": {
    "emotion_primary": "joy",
    "emotion_subtone": [
      "pride",
      "gratitude",
      "vitality"
    ],
    "higher_order_emotion": "triumphant resilience",
    "meta_emotional_state": "empowered",
    "interpersonal_affect": "belonging",
    "narrative_arc": "overcoming",
    "relational_dynamics": "community_ritual",
    "temporal_context": "midlife",
    "memory_type": "milestone",
    "media_format": "photo_with_story",
    "narrative_archetype": "hero",
    "symbolic_anchor": "stage performance",
    "relational_perspective": "community",
    "temporal_rhythm": "rising",
    "identity_thread": "dancer despite limits",
```

```
    "expressed_insight": "we all need something in our lives that we
can do just for us",
    "transformational_pivot": true,
    "somatic_signature": "mobile joints"
  },
  "governance": {
    "jurisdiction": null,
    "retention_policy": {
      "basis": "user_defined",
      "ttl_days": null,
      "on_expiry": "soft_delete"
    },
    "subject_rights": {
      "portable": true,
      "erasable": true,
      "explainable": false
    }
  },
  "telemetry": {
    "entry_confidence": 1,
    "extraction_model": "claude-sonnet-4-20250514"
  },
  "milky_way": {
    "event_type": "competition performance",
    "location_context": "physie competition venue",
    "associated_people": [
      "jason",
      "physie classmates"
    ],
    "visibility_context": "shared_publicly",
    "tone_shift": "celebratory reflection"
  },
  "gravity": {
    "emotional_weight": 0.8,
    "valence": "positive",
    "tether_type": "identity",
    "recurrence_pattern": "cyclical",
    "strength_score": 0.9
  }
}
```

Artifact B3 — Full Profile Artifact


```
{
  "meta": {
    "id": "c41c7878-48f4-40f2-a7ee-9817a2a837de",
    "version": "0.8.3",
    "profile": "full",
    "created_at": "2026-03-09T11:45:09.455Z",
    "updated_at": null,
    "locale": null,
    "owner_user_id": " vp-02HZ4GKWP7XTJY9QN7BN",
    "parent_id": null,
    "visibility": "private",
    "pii_tier": "moderate",
    "source_type": "text",
    "source_context": null,
    "consent_basis": "consent",
    "consent_scope": null,
    "consent_revoked_at": null,
    "tags": []
  },
  "core": {
    "anchor": "physie competition",
    "spark": "receiving competition photos",
    "wound": "body betrayal through arthritis",
    "fuel": "supportive women community",
    "bridge": "dance connecting past performer to present self",
    "echo": "feeling young on stage",
    "narrative": "Weekend physie competition photos arrived,
capturing the joy of performing despite psoriatic arthritis
diagnosis eight years ago. The supportive community of women and
weekly classes keep both joints mobile and spirits high. Dancing in
front of crowds resurrects the professional dancer within, defying
physical limitations. What began as Jason's encouragement four years
ago has become essential self-care for this working single mum."
  },
  "constellation": {
    "emotion_primary": "joy",
    "emotion_subtone": [
      "triumphant",
      "grateful",
      "empowered",
      "nostalgic"
    ],
    "higher_order_emotion": "self-actualization",
    "meta_emotional_state": "empowered",
    "interpersonal_affect": "belonging",
    "narrative_arc": "overcoming",
    "relational_dynamics": "community_ritual",
    "temporal_context": "midlife",
    "memory_type": "milestone",
```

```
    "media_format": "photo_with_story",
    "narrative_archetype": "hero",
    "symbolic_anchor": "dance stage",
    "relational_perspective": "community",
    "temporal_rhythm": "rising",
    "identity_thread": "dancer despite physical limitations",
    "expressed_insight": "we all need something in our lives that we
can do just for us",
    "transformational_pivot": true,
    "somatic_signature": "mobile joints"
  },
  "governance": {
    "jurisdiction": null,
    "retention_policy": {
      "basis": "user_defined",
      "ttl_days": null,
      "on_expiry": "soft_delete"
    },
    "subject_rights": {
      "portable": true,
      "erasable": true,
      "explainable": false
    },
    "exportability": "allowed",
    "k_anonymity": {
      "k": null,
      "groups": []
    },
    "policy_labels": [],
    "masking_rules": []
  },
  "telemetry": {
    "entry_confidence": 1,
    "extraction_model": "claude-sonnet-4-20250514",
    "extraction_provider": "anthropic",
    "extraction_notes": null,
    "alignment_delta": null
  },
  "milky_way": {
    "event_type": "competition",
    "location_context": "physie competition venue",
    "associated_people": [
      "Jason",
      "xt Physie Club",
      "Bxx Physie",
      "dance sisters"
    ],
    "visibility_context": "shared_publicly",
    "tone_shift": "struggle to triumph"
  },
  "gravity": {
```

```
    "emotional_weight": 0.8,
    "emotional_density": "high",
    "valence": "positive",
    "viscosity": "high",
    "gravity_type": "identity restoration",
    "tether_type": "identity",
    "recall_triggers": [
      "physie photos",
      "stage lights",
      "supportive cheers"
    ],
    "retrieval_keys": [
      "physie competition",
      "arthritis warrior",
      "dance community",
      "stage performance"
    ],
    "nearby_themes": [
      "chronic illness",
      "single motherhood",
      "self-care",
      "community support"
    ],
    "legacy_embed": true,
    "recurrence_pattern": "cyclical",
    "strength_score": 0.9,
    "temporal_decay": "slow",
    "resilience_markers": [
      "acceptance",
      "community",
      "adaptation"
    ],
    "adaptation_trajectory": "improving"
  },
  "impulse": {
    "primary_energy": "empowerment",
    "drive_state": "share",
    "motivational_orientation": "meaning",
    "temporal_focus": "present",
    "directionality": "outward",
    "social_visibility": "collective",
    "urgency": "elevated",
    "risk_posture": "bold",
    "agency_level": "high",
    "regulation_state": "regulated",
    "attachment_style": "secure",
    "coping_style": "seek_support"
  },
  "system": {
    "embeddings": [],
    "indices": {
```

```
      "waypoint_ids": [],
      "sector_weights": {
        "episodic": 0,
        "semantic": 0,
        "procedural": 0,
        "emotional": 0,
        "reflective": 0
      }
    }
  },
  "crosswalks": {
    "plutchik_primary": "joy",
    "geneva_emotion_wheel": null,
    "DSM5_specifiers": null,
    "HMD_v2_memory_type": "flashbulb",
    "ISO_27557_labels": null
  }
}
```

Artifact B4 — Partner Profile Artifact

(partner:com.deepadata.therapy.v1)

This artifact is extracted from the same source input as Artifacts B1–B3, under the `partner:com.deepadata.therapy.v1` partner profile. The base profile is Full (91 fields extracted and stored). The partner profile's `included_fields` manifest filters the API response to 24 clinically-focused affective fields. Comparing this artifact with Artifact B3 (Full Profile) demonstrates the two-lever cost model: the extraction is identical; only the response payload differs. The full manifest for this partner profile is defined in `schemas/partner-profiles/com.deepadata.therapy.v1.json` in the edm-spec repository.


```
{
  "meta": {
    "id": "f3a9c821-0d4e-4b7a-9f2e-8c1d5e6a3b77",
    "version": "0.8.3",
    "profile": "partner:com.deepadata.therapy.v1",
    "created_at": "2026-04-13T09:00:00Z",
    "locale": "en-au",
    "owner_user_id": "vp-01HXYZ",
    "visibility": "private",
    "pii_tier": "high",
    "source_type": "text",
    "source_context": "social media post",
    "consent_basis": "consent"
  },
  "core": {
    "anchor": "physie competition",
    "spark": "competition photos arriving",
    "wound": "psoriatic arthritis diagnosis",
    "bridge": "movement as ongoing defiance of limitation",
    "narrative": "Weekend physie competition photos arrived,
capturing the joy of performing despite psoriatic arthritis
diagnosis eight years ago. The supportive community of women and
weekly classes keep both joints mobile and spirits high. Dancing in
front of crowds resurrects the professional dancer within, defying
physical limitations."
  },
  "constellation": {
    "emotion_primary": "joy",
    "emotion_subtone": ["pride", "gratitude", "tenderness"],
    "narrative_arc": "overcoming",
    "relational_dynamics": "friendship",
    "identity_thread": "identity as performer and dancer persisting
beyond professional career and physical limitation",
    "arc_type": "liberation",
    "transformational_pivot": true,
```

```
    "somatic_signature": "joints mobile, body moving freely despite
chronic condition"
  },
  "gravity": {
    "emotional_weight": 9,
    "tether_type": "community",
    "recurrence_pattern": "weekly",
    "strength_score": 8,
    "temporal_decay": "low",
    "resilience_markers": ["continues despite chronic illness",
"adapted dance form", "maintained identity"],
    "adaptation_trajectory": "stable"
  },
  "impulse": {
    "regulation_state": "regulated",
    "agency_level": "high",
    "coping_style": "active",
    "attachment_style": "secure"
  }
}
```


Note: Meta, Governance, Telemetry, System, and Crosswalks domains are omitted from this response per the `included_fields` manifest. They are present in the stored base-profile artifact. Milky_Way fields (event_type, location_context, etc.) are present in the stored Full artifact but excluded from the therapy partner profile response by design — contextual framing fields are not clinically essential for session continuity.

Appendix C — The .ddna Envelope Format (v1.1)

The **.ddna** envelope is the cryptographically anchored, governance-preserving container that binds an EDM artifact into a durable, portable, and sovereign emotional memory unit.

C.1 Minimum Normative Envelope Structure

Below is the minimal valid structure for a compliant .ddna v1.1 envelope containing an EDM v0.8.3 artifact.

```
{
  "ddna_header": {
    "ddna_version": "1.1",
    "created_at": "2026-01-15T10:00:00Z",
    "edm_version": "0.8.3",
    "owner_user_id": " vp-02HZ4GKWP7XTJY9QN7BN",
    "exportability": "allowed",
    "jurisdiction": "AU",
    "retention_policy": {
      "basis": "user_defined",
      "ttl_days": null,
      "on_expiry": "soft_delete"
    },
    "consent_basis": "explicit_consent",
    "masking_rules": [],
    ~~"payload_type": "edm.v0.6.0",~~ "payload type": "edm.v0.8.3",
    "audit_chain": [
      {
        "at": "2026-01-15T10:00:00Z",
        "event": "created",
        "agent": "ddna-tools/0.1.0"
      }
    ]
  },
  "edm_payload": {
    "meta": {
      "id": "c3e1d6a7-8f2b-410a-9d3c-5e6f7a8b9c0d",
      "version": "0.8.3",
      "profile": "full",
      "..." : "..."
    },
    "core": { "...": "..." },
    "constellation": { "...": "..." },
    "milky_way": { "...": "..." },
    "gravity": { "...": "..." },
    "impulse": { "...": "..." },
    "governance": { "...": "..." },
    "telemetry": { "...": "..." },
    "system": { "...": "..." },
```

```
    "crosswalks": { "...": "..." }
  },
  "proof": {
    "type": "DataIntegrityProof",
    "cryptosuite": "eddsa-jcs-2022",
    "created": "2026-01-15T10:00:00Z",
    "verificationMethod":
"did:key:z6MkiTBz1ymuepAQ4HEHYSF1H8quG5GLVVQR3djdX3mDooWp",
    "proofPurpose": "assertionMethod",
    "proofValue":
"z3FXQeKviyRpKzPHgXtUfB3GRdJHWbvEAiDMYPJ7vLxQCdDmtJUnXfJNS4R8kPJF5w.
.."
  }
}
```

C.2 Header Fields

The `ddna_header` contains the following fields:

Field	Type	Description
`ddna_version`	string	Envelope format version. **MUST** be "1.1" for this specification.
`created_at`	string	ISO-8601 UTC timestamp of envelope creation.
`edm_version`	string	EDM schema version (e.g., "0.8.3").
`owner_user_id`	string | null	Vitapass or vendor-specific subject identifier. Null for stateless.
`exportability`	string	One of: `allowed`, `restricted`, `prohibited`.
`jurisdiction`	string	Regulatory jurisdiction code (e.g., "AU", "GDPR", "CCPA").
`retention_policy`	object	Contains `basis`, `ttl_days`, `on_expiry`.
`consent_basis`	string	Legal basis for processing.
`masking_rules`	array	Redaction requirements for display or export.
`payload_type`	string	Type identifier for the payload (e.g., "edm.v0.8.3").
`audit_chain`	array	Lifecycle events. Each entry has `at`, `event`, `agent`.

Note: The `encryption` and `compression` fields present in v1.0 have been removed. These concerns are delegated to transport and storage layers.

C.3 Proof Block (W3C Data Integrity)

The `proof` block follows the W3C Data Integrity specification using the `eddsa-jcs-2022` cryptosuite:

Field	Type	Description
`type`	string	**MUST** be "DataIntegrityProof".
`cryptosuite`	string	**MUST** be "eddsa-jcs-2022".
`created`	string	ISO-8601 UTC timestamp when the proof was created.
`verificationMethod`	string	DID URL identifying the signing key.
`proofPurpose`	string	**MUST** be "assertionMethod".
`proofValue`	string	Multibase base58-btc encoded signature (prefix `z`).
`previousProof`	string	Optional. Reference to a prior proof, used for proof chains in multi-party signing scenarios.
`expires`	string	Optional. ISO-8601 timestamp after which proof is invalid.
`domain`	string	Optional. Domain binding for proof scope.
`challenge`	string	Optional. Challenge value for proof freshness.
`nonce`	string	Optional. Nonce for replay protection.

C.4 Signing Input Construction

The signing input is computed per the W3C Data Integrity specification:

1. Create `proof_options`: The proof block without `proofValue`
2. Create `document`: `{ "ddna_header": ..., "edm_payload": ... }`
3. Canonicalize both using RFC 8785 JCS
4. Compute: `signing_input = SHA-256(JCS(proof_options)) || SHA-256(JCS(document))`
5. Sign with Ed25519: `signature = Ed25519.sign(signing_input, private_key)`
6. Encode: `proofValue = multibase_base58btc(signature)`

C.5 Verification Methods

The `verificationMethod` field contains a DID URL that resolves to the signing public key:

- `did:key:z6Mk...` — Self-contained Ed25519 public key (recommended for portability)
- `did:web:example.com#key-1` — Web-resolvable DID document
- `did:vitapass:...` — Future VitaPass-native DID method

C.6 Audit Chain

The `audit_chain` array records lifecycle events. Each entry contains:

Field	Type	Description
`at`	string	ISO-8601 UTC timestamp of the event.
`event`	string	Event type: `created`, `transferred`, `verified`, `expired`
`agent`	string	Identifier of the system or actor that performed the event.
`details`	String	Optional. Additional context for the event — e.g., the agent version, trigger source, or migration notes. Free text.

Audit chain entries are append-only. The chain is signed as part of the header, so modifications after sealing are detectable. Systems **MAY** append entries to the audit chain after sealing for tracking purposes, but such additions invalidate the original proof and require re-sealing.

C.7 Migration from v1.0

Envelopes created under v1.0 with `ddna_integrity` blocks remain valid and interpretable. To migrate:

1. Remove `ddna_integrity` block (which contained `payload_hash`, `header_hash`, `signature`, `signed_by`, and `signature_algorithm` fields)
2. Remove `encryption` and `compression` fields from header (if present)
3. Add `audit_chain` to header (migrate from `ddna_integrity.audit_chain` if it was present)
4. Update `ddna_version` to "1.1"
5. Create and attach W3C Data Integrity `proof` block
6. Re-seal with appropriate signing key

Appendix D — Validation Summary

Protocol-level requirements for representational and persistence validity.

Validation ensures that EDM artifacts maintain structural integrity, interpretive stability, and governance continuity across systems, vendors, and temporal horizons. It is the mechanism through which the representational and persistence layers enforce the guarantees defined in the core architecture. Validation is not an implementation detail; it is a protocol obligation that constrains how artifacts may be created, transported, interpreted, and sealed.

Validation applies independently to the representational layer (EDM schema) and the persistence layer (.ddna envelope). An artifact is valid only when all representational requirements are satisfied and all persistence constraints, when applicable, are met. Invalid artifacts **MUST NOT** be admitted into any computational workflow. Invalid artifacts **MUST NOT** be encapsulated within a .ddna envelope. All interpretive and computational behaviour is downstream of validation.

D.1 Representational Validity

An EDM artifact is representationally valid when its structure and content conform to all mandatory schema constraints. These requirements ensure that every artifact produced under the specification remains fully interpretable, non-inferential, and profile-complete.

Representational validity requires the following conditions:

1. Domain Presence

All domains declared by the artifact's implementation profile **MUST** be present. Out-of-profile domains **MUST** be omitted entirely. Fields within declared domains may contain explicit null values but **MUST NOT** be omitted.

2. Schema Alignment

All fields defined by the canonical schema **MUST** be initialised and **MUST** conform to their declared type, permitted value sets, and canonical enumerations. Temporal fields **MUST** use ISO-8601 UTC format. Categorical values **MUST** use canonical token forms. Embedding references **MUST** be indirections; no inline vectors are permitted.

3. Non-Inferential Semantics

All populated values **MUST** originate from explicit narrative content, observable structure, user-declared metadata, or deterministic extraction logic. Latent activations, psychological constructs, behavioural predictions, or unverifiable model-internal states **MUST NOT** enter the representational payload.

4. Governance Completeness

When identity-bearing or persistence-relevant fields are present, the `governance` domain **MUST** be complete, internally consistent, and unambiguous. Governance fields **MUST** be declared in accordance with the artifacts profile.

5. Deterministic Serialisation

The artifact **MUST** serialise deterministically. No extraneous fields, unbounded structures, or order-dependent ambiguities are permitted. Byte-stable identity **MUST** be possible from the serialised form.

An artifact failing any representational requirement is invalid.

D.2 Persistence Validity (.ddna Envelope)

A .ddna envelope is valid only when its structural, cryptographic, and governance requirements are satisfied. Persistence validity is distinct from representational validity: an artifact may be representationally valid but ineligible for sealing if persistence constraints are not met.

Persistence validity requires the following conditions:

1. Envelope Structure

The envelope **MUST** contain a header (`ddna_header`), proof block (`proof`), and payload (`edm_payload`) in the arrangement defined by this specification.

2. Integrity Guarantees

The envelope **MUST** contain a valid W3C Data Integrity proof. The proof **MUST** verify against the canonical form of the header and payload using the specified verification method. The `cryptosuite` **MUST** be `eddsa-jcs-2022`. The `proofPurpose` **MUST** be `assertionMethod`.

3. Payload Conformity

The EDM artifact referenced by the envelope **MUST** be representationally valid, **MUST** appear exactly as hashed, and **MUST NOT** differ in any respect from the payload referenced by the integrity block.

4. Governance Enforcement

Envelope creation **MUST** reflect and enforce all governance constraints of the payload. Exportability, retention, jurisdiction, consent, and masking declarations **MUST** be honoured at sealing. If exportability prohibits persistence, sealing is prohibited.

5. Byte-Stable Sealing

A sealed envelope **MUST** maintain byte-stable identity across storage and transport. Any modification of header, integrity block, or payload invalidates the envelope.

An envelope failing any persistence requirement is invalid.

D.2.1 Verification Order Requirements

Systems processing .ddna envelopes **MUST** follow this verification sequence:

1. **Load**: Read envelope from storage or transport
2. **Parse**: Deserialise JSON structure
3. **Structural Check**: Verify presence of `ddna_header`, `edm_payload`, `proof`
4. **Proof Verification**: Verify cryptographic proof BEFORE accessing payload content
    a. Resolve `verificationMethod` to obtain public key
    b. Reconstruct signing input from canonical forms
    c. Verify Ed25519 signature
5. **Governance Check**: Evaluate `exportability`, `visibility`, retention status
6. **Content Access**: Only after all checks pass, expose payload to consumer

Systems **MUST NOT** expose envelope content before completing proof verification. This ordering prevents processing of tampered or forged envelopes.

D.2.2 MIME Type Handling

Systems handling .ddna files **MUST** recognise the following MIME types:

MIME Type	**Usage**
`application/json`	Unsealed EDM artifacts (`edm.json`)
`application/vnd.deepadata.ddna+json`	Sealed .ddna envelopes (`.ddna`)

When serving or receiving .ddna files over HTTP or other protocols, the `application/vnd.deepadata.ddna+json` MIME type signals that the content is a governed envelope requiring signature verification.

D.3 Validator Responsibilities

A conformant validator **MUST** enforce all representational and persistence requirements defined above. Validators operate at the protocol boundary and determine whether an artifact may participate in representational, persistence, or computational workflows.

Validator responsibilities include:

1. Pre-Admission Enforcement

No artifact may be admitted into any computational workflow until representational validity is confirmed.

2. Pre-Sealing Enforcement

No artifact may enter the .ddna sealing workflow unless both representational and persistence validity conditions are satisfied.

3. Non-Modification Requirement

Validators **MUST NOT** modify, infer, reconstruct, or reinterpret representational content. The only permissible normalisation is deterministic structural normalisation that does not alter schema semantics, value meaning, or governance declarations.

4. Failure Handling

Artifacts or envelopes failing any requirement are invalid and **MUST NOT** be processed, persisted, or transported.

5. Interpretive Isolation

Validation outcomes **MUST NOT** alter or introduce representational semantics. Validation determines admissibility, not interpretive meaning.

D.4 Stateless Operation

Stateless operation corresponds to the architectural path in which a fully formed EDM artifact is used exclusively within an immediate computational session and is not committed to sovereign persistence. The extraction engine produces a complete, schema-valid EDM artifact. The commit decision determines whether that artifact enters the .ddna sealing workflow (Path B) or remains confined to stateless coherence (Path A).

A stateless artifact is a complete EDM representation derived from the non-commit outcome of this decision. Stateless artifacts remain fully subject to all representational validity requirements defined in D.1. Statelessness does not alter domain composition; it constrains the semantics of selected domains to ensure that no identity-bearing or persistence-relevant information is available beyond the active session.

The following invariants define stateless operation:

Identity Neutrality

Fields that bind the artifact to a persistent, real-world, or account-level identity **MUST NOT** contain identifying values in the retained stateless representation. Structural positions remain unchanged, but identity resolution is prohibited.

Context Neutrality

The `milky_way` domain **MUST** remain structurally present in profiles that declare it to satisfy profile completeness. In profiles that do not declare `milky_way` (Essential), the domain is absent entirely. However, in a stateless artifact where `milky_way` is declared, all fields within the domain **MUST** be set to null.

This guarantees that the stateless representation carries zero persistent or re-identifiable context while preserving the structural position of the domain.

Salience Neutrality

The `gravity` domain **MUST** remain structurally present in profiles that declare it to satisfy profile completeness. In profiles that do not declare `gravity` (Essential), the domain is absent entirely. However, in a stateless artifact where `gravity` is declared, all fields within the domain **MUST** be set to null."

This ensures that the stateless representation carries no persistence, prioritisation, or longitudinal salience semantics of any form.

Governance Neutrality

The `governance` domain **MUST** indicate non-persistence. Retention, exportability, and sovereignty semantics **MUST NOT** permit archival storage, inheritance, or cross-system portability. Stateless artifacts cannot become sovereign emotional memory objects.

Telemetry and System Neutrality

The `telemetry` and `system` domains **MUST NOT** contain identifiers, statistics, or linkage structures implying continuity across sessions. No retrieval-history or long-horizon usage signals MAY be encoded.

Persistence Prohibition

Stateless artifacts are ineligible for sealing. .ddna envelopes **MUST NOT** be created from stateless artifacts. No persistence semantics apply.

Stateless operation preserves full representational fidelity during the session while ensuring that artifacts not committed to sovereign persistence cannot function as rights-bearing or persistently identifiable emotional records.

D.5 Summary

Validation enforces the structural, semantic, and governance guarantees upon which EDM interoperability depends. Representational validity ensures that the artifact conforms to the schema and contains only explicit, auditable, non-inferential content. Persistence validity ensures that the envelope maintains integrity, sovereignty, and transport safety. Stateless operation restricts persistence, not representation. Validators act as architectural gatekeepers, ensuring that only conformant artifacts participate in representational, persistence, and computational workflows.

This appendix defines the minimal normative validation contract for EDM v0.8.3. Extended validator profiles, reference implementations, and operational guidelines MAY be maintained in separate companion documentation.

D.6 Reference Implementation

The `ddna-tools` package provides a reference implementation of the sealing and verification procedures described in this specification. Test vectors and canonical examples are maintained in the tool repository.

Appendix E — Glossary of Protocol Terms

This glossary defines all terms used in the EDM v0.8.3 representational schema, the .ddna persistence layer, and the Emotional Data Architecture Stack. All definitions are sourced directly from the specification and are free of behavioural or inferential semantics.

A

Affective Context The structured emotional content extracted from input and encoded explicitly into EDM fields. Never inferred from latent states.

Affective Manifold The domains declared by the artifact's implementation profile.

Anchor A Core primitive identifying the central person, idea, object, or symbolic focus of an experience.

Audit Chain The append-only array in the ddna_header recording lifecycle events (created, transferred, verified, expired). Signed as part of the envelope; modifications after sealing invalidate the proof.

B

Bridge A Core primitive describing the cognitive or emotional shift that allows reframing, integration, or reconciliation within the narrative.

Byte-Stable A property of serialized artifacts requiring that the payload remains bit-for-bit identical across storage systems, enabling deterministic hashing and long-term interpretability.

C

Canonical Enumerations Normative sets of permitted values defined by the EDM schema. Vendors MUST NOT modify or extend these sets without versioning.

Compute Boundary The architectural line separating the representational and persistence layers (EDM and .ddna) from all downstream computation.

Constellation The EDM domain encoding emotional topology, including primary emotion, subtones, archetype, higher-order emotion, meta-emotional state, insight, and related descriptors.

Consent Basis A Governance descriptor specifying the legal foundation for storing or processing affective data (e.g., consent, contract).

Core Anchor, spark, wound, fuel, bridge, and echo. Extended and Full profiles additionally include narrative.

Crosswalks A referential mapping to external taxonomies (e.g., Plutchik, ISO) that does not alter EDM semantics.

D

.ddna Envelope The sovereign, cryptographically anchored persistence container that encapsulates an EDM payload with governance rules, version anchoring, and integrity guarantees.

Domain Completeness This term is superseded by Profile Completeness in v0.8.0..

Data Integrity Proof A W3C standard cryptographic proof structure that binds a signature to a document using a specified cryptosuite.

DID (Decentralized Identifier) A URI scheme for verifiable, decentralized digital identity. Used in `verificationMethod` fields.

Declared Domain A domain explicitly included in an implementation profile's field set. Declared domains must be present in conformant artifacts. Out-of-profile domains must be omitted entirely; they may not be present with null values.

E

Echo A Core primitive describing the residual continuity or after-image of an emotional experience.

EDM (Emotional Data Model) A schema-bound representational architecture externalizing affective context as explicit, model-neutral data.

Embeddings (System Domain) Stable external references to embedding systems (provider, sector, dim, vector_ref). Vectors themselves MUST NOT be embedded within EDM artifacts.

Explicit Representation The principle requiring that all fields encode only extracted or declared content. No latent reconstruction or psychological inference is permitted.

Exportability A Governance descriptor determining whether an EDM artifact may be exported as a .ddna object.

Expressed Insight A Constellation field capturing the explicit realization stated by the subject. Extracted directly from input.

eddsa-jcs-2022 The W3C Data Integrity cryptosuite used for .ddna envelope signing. Combines Ed25519 signatures with JCS canonicalization.

F

Fuel A Core primitive capturing the positive value, driver, or attachment that energizes the emotional state.

G

Governance (Domain) The in-band metadata layer encoding jurisdiction, consent, retention, exportability, subject rights, and policy labels.

Gravity The salience-dynamics domain capturing emotional weight, density, valence, viscosity, tethering, decay, and recurrence patterns.

H

Higher-Order Emotion A Constellation descriptor representing complex or layered emotional states beyond primary or secondary affect.

I

Identity Thread A Constellation field describing how the experience relates to or shapes the subject's personal identity.

Implicit vs Explicit Affect Implicit (model-internal, latent) affect is prohibited. Explicit affect must be encoded as structured EDM fields.

Impulse The domain capturing motivational vectors including drive_state, urgency, directionality, regulation_state, social visibility, and agency.

In-Band Governance The architectural principle requiring governance metadata to exist within the artifact itself.

Integrity Block (.ddna) A cryptographic structure storing hashes and optional signatures for tamper detection and validation.

Implementation Profile One of three tiers — Essential, Extended, or Full — defining the exact set of domains and fields an artifact must contain. Profile conformance is enforced at validation. See Section 3.7.

J

Jurisdiction A Governance descriptor indicating the regulatory domain applicable to rights, retention, and compliance.

JCS (JSON Canonicalization Scheme) RFC 8785 standard for deterministic JSON serialization. Required for cryptographic signing.

K

K-Anonymity A Governance structure specifying anonymization rules for safe aggregation or export.

L

Legacy Embed A Gravity indicator describing whether the experience contributes to long-range personal identity structure.

Locale A Meta descriptor identifying language and regional formatting using BCP47 codes.

M

Memory Type A Constellation descriptor defining the experiential category (milestone, reflection, legacy_artifact, etc.).

Meta The EDM domain containing identifiers, timestamps, source metadata, visibility, PII tier, session, and provenance.

Milky_Way The contextual framing domain: event type, location, media, relational roles, tone shifts, and contextual layers.

Model Neutrality An EDM design principle ensuring independence from any specific embedding, architecture, or inference system.

Multibase An encoding scheme with a prefix character indicating the base encoding. The .ddna specification uses base58-btc (prefix `z`).

N

Narrative A compressed, high-signal semantic summary (3–5 sentences) representing the structural essence of the experience. For Extended and Full profiles.

Narrative Archetype A Constellation descriptor assigning the archetypal identity the subject embodies (hero, seeker, caregiver, sage, etc.); an identity, not a structural role.

Non-Inferential Representation A representational constraint prohibiting inference of psychological or behavioral states.

O

Owner_user_id A Meta field linking the artifact to a human subject. MUST be null in Stateless Mode.

P

Payload (.ddna) The serialized EDM artifact preserved bit-for-bit within the envelope.

PII Tier A Meta descriptor classifying personal data sensitivity.

Policy Labels Governance tags used for applying additional constraints.

Proof Block The W3C Data Integrity proof structure in a .ddna envelope, containing signature and verification metadata.

Profile Completeness The v0.8.0 invariant requiring that all domains declared by an artifact's implementation profile are present, and all out-of-profile domains are absent. Supersedes Domain Completeness.

Proof Chain A sequence of linked cryptographic proofs enabling multi-party signing of a .ddna envelope. Each proof references the prior via the `previousProof` field.

R

Recurrence Pattern A Gravity field describing the observed temporal repetition structure of an emotional pattern.

Representational Stability The guarantee that EDM artifacts remain interpretable across computational eras and model updates.

Retention Policy A Governance descriptor defining time-to-live and expiry actions.

S

Salience (Gravity) The structural importance of an emotional experience as encoded by weight, density, and tethering.

Sealed Artifact An EDM artifact encapsulated within a .ddna envelope under full governance and integrity constraints.

Somatic Signature A Constellation field encoding explicit interoceptive descriptors present in the input text.

Spark A Core primitive describing the initiating perceptual or emotional trigger.

Stateless Configuration A reduced-governance mode in which domains declared by the profile that carry contextual or salience semantics **MUST** be nulled.

Stateless Artifact A complete, schema-valid EDM artifact produced when the commit decision results in non-persistence. Subject to all representational validity requirements but ineligible for sealing. Identity, contextual, and salience domains are neutralised per D.4.

System (Domain) The domain encoding embedding system references, retrieval indices, and sector weights.

T

Telemetry A domain storing extraction confidence, model ID, extraction notes, and alignment delta.

Tether Type / Tether Target Gravity descriptors identifying the anchor category and specific anchor object.

Temporal Context / Temporal Rhythm Constellation descriptors capturing framing in time and recurrence cadence.

U

Urgency An Impulse field representing immediacy or intensity of motivational state.

V

Valence A Gravity descriptor representing emotional direction (positive, negative, mixed).

Vector_ref A System-domain pointer to externally stored embedding vectors.

Verification Method A DID URL in the proof block that resolves to the public key used for signature verification.

VitaPass A sovereign identity and consent infrastructure that enables portable, cross-vendor identity resolution while preserving strict separation between identity metadata and affective representation.

W

Wound A Core primitive describing the activated vulnerability, loss, or unresolved theme present in the emotional experience.